



Linear Algebra

Centennial College SETAS Custom Text

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Unit 1: Systems of Linear Equations

1.1 - Introduction to Gaussian Elimination

A **system of linear equations** is a finite set of linear equations.

To solve a system of linear equations means to determine the set of values that satisfies all the equations in that linear system.

For example, given the following system of linear equations:

$$x + y = 5$$

$$2x - 3y = -5$$

The solution of the linear system would be $x=2$ and $y=3$ because this set of values satisfies both the equations.

There are various methods to solve a linear system. Substitution and Elimination are both useful methods. In this chapter, Gaussian Elimination is introduced as a new method, in which matrices will be used in the process.

A **matrix**, which is a rectangular array of numbers, needs to be created to begin the process of Gaussian Elimination. Using the coefficients of x and y and the constant in each equation, the above linear system is written as

$$\begin{bmatrix} 1 & 1 & 5 \\ 2 & -3 & -5 \end{bmatrix}$$

The concept behind the method of Gaussian Elimination is similar to the concept behind performing Elimination. The two methods are shown in parallel:

Elimination:

$$\begin{array}{r} x + y = 5 \\ 2x - 3y = -5 \end{array}$$

Step 1:

Multiply equation 1 by 2

$$\begin{array}{r} 2x + 2y = 10 \\ 2x - 3y = -5 \end{array}$$

Step 2:

Take equation 1 subtract equation 2
so that x would be eliminated

$$5y = 15$$

Step 3:

Gaussian Elimination:

$$\begin{bmatrix} 1 & 1 & 5 \\ 2 & -3 & -5 \end{bmatrix}$$

Step 1:

Multiply row 1 (R1) by 2

$$\begin{bmatrix} 2 & 2 & 10 \\ 2 & -3 & -5 \end{bmatrix}$$

Step 2:

Take row 1 (R1) subtract row 2 (R2)

$$\begin{bmatrix} 2 & 2 & 10 \\ 0 & 5 & 15 \end{bmatrix}$$

Step 3:

Solve for y from the above step

$$y = \frac{15}{5}$$
$$y = 3$$

Step 4:

Using y , solve for x

$$x + 3 = 5$$
$$x = 2$$

Divide R_2 by 5

$$\begin{bmatrix} 2 & 2 & 10 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 4:

Divide R_1 by 2

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 5:

Rewrite R_1 by taking R_1 subtract R_2

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$$

Both methods resulted in the solution of $x=2$ and $y=3$

From the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$, this represents,

$$1x + 0y = 2$$

$$0x + 1y = 3$$

Which gave the solutions $x=2$ and $y=3$.

Details about the method of Gaussian Elimination

At each step of the process of Gaussian Elimination, an operation is performed on a row. These are known as **elementary row operations**, which includes the following:

- 1) Multiplying an entire row by a non-zero constant
- 2) Interchanging two rows
- 3) Adding or subtracting two rows

In some cases, operation 1 and 3 can be combined as one step. For example, multiply R_1 by 3 then subtract R_2 .

The goal of performing elementary row operations is to determine a matrix in **reduced row-echelon form**, which has the following properties:

- 1) The first non-zero entry from the left of the matrix is called a **leading 1**.
- 2) Each leading 1 is to the right of all leading 1's in the rows above it.
- 3) If there are rows that consists entirely of zeros, they are placed at the bottom of the matrix.
- 4) Each leading 1 is the only non-zero entry in its column.

In the above example, performing elementary row operations carried the matrix from

$$\begin{bmatrix} 1 & 1 & 5 \\ 2 & -3 & -5 \end{bmatrix} \text{ to } \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$$

The matrix, $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$, is a matrix in reduced row-echelon form that gave the solutions of the linear system.

Example 1

Using Gaussian Elimination, solve the given linear system

$$2x + 2y = 14$$

$$4x - 3y = 7$$

Step 1: Create a matrix

$$\begin{bmatrix} 2 & 2 & 14 \\ 4 & -3 & 7 \end{bmatrix}$$

Step 2: Rewrite R2 by taking 2 multiply by R1, then subtract R2

$$\begin{bmatrix} 2 & 2 & 14 \\ 0 & 7 & 21 \end{bmatrix}$$

Step 3: Rewrite R1 by dividing by 2

$$\begin{bmatrix} 1 & 1 & 7 \\ 0 & 7 & 21 \end{bmatrix}$$

Now a leading 1 and a zero entry along its column is created.

The next operations will be carried out to create a leading 1 in R2 and a zero entry along its column.

Step 4: Rewrite R2 by dividing by 7

$$\begin{bmatrix} 1 & 1 & 7 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 5: Rewrite R1 by subtracting R2 from R1

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix}$$

At this point, the matrix is now in reduced row-echelon form.

Therefore, the solutions are $x = 4$ and $y = 3$

Example 2

Using Gaussian Elimination, solve the given linear system

$$4a + b + 1 = 0$$

$$a - \frac{13}{2} = 2b$$

Step 1: Rewrite the equations

$$4a + b = -1$$

$$a - 2b = \frac{13}{2}$$

Step 2: Create the matrix

$$\begin{bmatrix} 4 & 1 & -1 \\ 1 & -2 & \frac{13}{2} \end{bmatrix}$$

Step 3: Interchange $R1$ and $R2$

$$\begin{bmatrix} 1 & -2 & \frac{13}{2} \\ 4 & 1 & -1 \end{bmatrix}$$

Step 4: Rewrite $R2$ by taking $R1$ multiply by 4, then subtract $R2$

$$\begin{bmatrix} 1 & -2 & \frac{13}{2} \\ 0 & -9 & 27 \end{bmatrix}$$

Step 5: Rewrite $R2$ by dividing by -9

$$\begin{bmatrix} 1 & -2 & \frac{13}{2} \\ 0 & 1 & -3 \end{bmatrix}$$

Step 6: Rewrite $R1$ by adding $R2$ with 2 times $R2$

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -3 \end{bmatrix}$$

The solution of the linear system is $x = \frac{1}{2}$ and $y = -3$

Example 3

Using Gaussian Elimination, solve the given linear system

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$R_2' = 2R_1 - R_2$$

(where R_2' represents the new R_2)

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & -2 & 7 & 17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$R_3' = 3R_1 - R_3$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & -2 & 7 & 17 \\ 0 & -3 & 11 & 27 \end{bmatrix}$$

$$R_2' = R_2 \div -2$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -3.5 & -8.5 \\ 0 & -3 & 11 & 27 \end{bmatrix}$$

$$R_1' = R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 5.5 & 12.5 \\ 0 & 1 & -3.5 & -8.5 \\ 0 & -3 & 11 & 27 \end{bmatrix}$$

$$R_3' = 3R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & 5.5 & 12.5 \\ 0 & 1 & -3.5 & -8.5 \\ 0 & 0 & 0.5 & 1.5 \end{bmatrix}$$

$$R3' = R3 \times 2$$

$$\begin{bmatrix} 1 & 0 & 5.5 & 12.5 \\ 0 & 1 & -3.5 & -8.5 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$R2' = R2 + (R3 \times 3.5)$$

$$\begin{bmatrix} 1 & 0 & 5.5 & 12.5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$R1' = R1 - (R3 \times 5.5)$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Practice

Solve the following systems of linear equation using Gaussian Elimination

$$\begin{aligned} 1) \quad & 3x + 5y = 31 \\ & 2x + 3y = 20 \end{aligned}$$

$$\begin{aligned} 2) \quad & 74 + 5x = -3y \\ & -3y = 26 + 2x \end{aligned}$$

$$\begin{aligned} 3) \quad & y = -x + 2 \\ & y = 4x - 3 \end{aligned}$$

$$\begin{aligned} 4) \quad & a + 2b + c = -3 \\ & 2a - b + 3c = 3 \\ & -a + 5b + 5c = -11 \end{aligned}$$

$$\begin{aligned} 5) \quad & 3x + 4y + z = 1 \\ & 2x + 3y = 0 \\ & 4x + 3y - z = -2 \end{aligned}$$

$$\begin{aligned} 6) \quad & 2x + 1 = -y - z \\ & x + 2y = -z \\ & 3x = 5 + 2z \end{aligned}$$

1.2 - Types of Solutions for Linear Systems

When solving linear systems, there are three types of solutions that are possible:

- 1) One unique solution
- 2) No solution
- 3) Infinitely many solutions

Example 1

From Chapter 1.1, the following linear system had **one unique solution**:

$$x + y = 5$$

$$2x - 3y = -5$$

After performing Gaussian Elimination, the matrix became $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$, resulting in one unique solution:

$$x = 2 \text{ and } y = 3$$

Example 2

The following linear system has **no solutions**, in other words, the system is **inconsistent**

$$9x + 8y = -8$$

$$54x + 48y = -47$$

Performing Gaussian Elimination,

$$\begin{bmatrix} 9 & 8 & -8 \\ 54 & 48 & -47 \end{bmatrix}$$

$$R_2' = 6R_1 - R_2$$

$$\begin{bmatrix} 9 & 8 & -8 \\ 0 & 0 & -1 \end{bmatrix}$$

$$R_1' = R_1 \div 9$$

$$\begin{bmatrix} 1 & \frac{8}{9} & \frac{-8}{9} \\ 0 & 0 & -1 \end{bmatrix}$$

This matrix represents

$$x + \frac{8}{9}y = \frac{-8}{9}$$

$$0x + 0y = -1$$

There are no values of x and y that will satisfy the equation $0x + 0y = -1$

Therefore, in general, if a row with $[0 \ 0 \ \dots \ 0 \ k]$ appears, where k is a non-zero value, then the linear system has no solutions.

Example 3

The following linear system has **infinitely many solutions**:

$$5x + 6y = 13$$

$$15x + 18y = 39$$

Performing Gaussian Elimination,

$$\begin{bmatrix} 5 & 6 & 13 \\ 15 & 18 & 39 \end{bmatrix}$$

$$R_2' = 3R_1 - R_2$$

$$\begin{bmatrix} 5 & 6 & 13 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R1' = R1 \div 5$$

$$\begin{bmatrix} 1 & \frac{6}{5} & \frac{13}{5} \\ 0 & 0 & 0 \end{bmatrix}$$

This matrix represents

$$x + \frac{6}{5}y = \frac{13}{5}$$

$$0x + 0y = 0$$

Any value of x and y would satisfy the equation

$$0x + 0y = 0$$

Therefore, this linear system has infinitely many solutions.

In general, if at least one row of zero appears, then the linear system has infinitely many solutions.

When a linear system has infinitely many solutions, **parametric equations** are used to represent the solution.

From the corresponding linear system, since there are infinite values of y that satisfies $0x + 0y = 0$

A parameter is assigned to the y variable. Let $y = r$

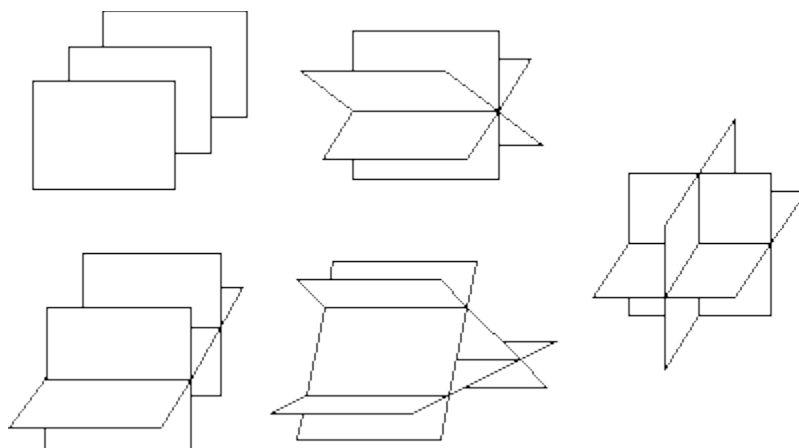
Then writing x in terms of the parameter will result in $x = \frac{13}{5} - \frac{6}{5}r$

So the solution of the linear system is:

$$x = \frac{13}{5} - \frac{6}{5}r$$

$$y = r$$

From the above examples, each of the three cases can be visualized as the following:



From the top left corner, going clockwise:

- 1) No solutions, the three planes do not intersect
- 2) Infinite solutions, the three planes intersect at a line, which can be represented by parametric equations
- 3) One solution, the three planes intersect at a single point
- 4) No solutions, the three planes do not intersect at a common point or line
- 5) No solutions, the three planes do not intersect at a common point or line

Example 4

Determine the solution of the given linear system.

$$x + 2y - 3z = -3$$

$$x + 3y - 5z = 5$$

$$x - 2y + 5z = -35$$

$$\begin{bmatrix} 1 & 2 & -3 & -3 \\ 1 & 3 & -5 & 5 \\ 1 & -2 & 5 & -35 \end{bmatrix}$$

$$R_2' = R_1 - R_2$$

$$\begin{bmatrix} 1 & 2 & -3 & -3 \\ 0 & -1 & 2 & -8 \\ 1 & -2 & 5 & -35 \end{bmatrix}$$

$$R_3' = R_1 - R_3$$

$$\begin{bmatrix} 1 & 2 & -3 & -3 \\ 0 & -1 & 2 & -8 \\ 0 & 4 & -8 & 32 \end{bmatrix}$$

$$R2' = R2 \div -1$$

$$\begin{bmatrix} 1 & 2-3 & -3 \\ 0 & 1-2 & 8 \\ 0 & 4-8 & 32 \end{bmatrix}$$

$$R1' = R1 - (R2 \times 2)$$

$$\begin{bmatrix} 1 & 0 & 1 & -19 \\ 0 & 1-2 & 8 \\ 0 & 4-8 & 32 \end{bmatrix}$$

$$R3' = 4 R2 - R3$$

$$\begin{bmatrix} 1 & 0 & 1 & -19 \\ 0 & 1-2 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

The corresponding linear system is:

$$x + z = -19$$

$$y - 2z = 8$$

$$0z = 0$$

Let $z = r$

Then the solution of the linear system is:

$$x = -19 - r$$

$$y = 8 + 2r$$

$$z = r$$

Practice

- 1) Solve the following systems of linear equations. State the type of solution.
Where appropriate, use parametric equations to represent the solution.

$$\begin{aligned} \text{a) } & 3x - y = 0 \\ & 2x - 3y = 1 \end{aligned}$$

$$\begin{aligned} \text{b) } & 2x - 3y = 5 \\ & 3y - 2x = 2 \end{aligned}$$

$$\begin{aligned} \text{c) } & 10x - 2y = -3 \\ & -5x + y = 1 \end{aligned}$$

$$\begin{aligned} \text{d) } -2x + 3y + 3z &= -9 \\ 3x - 4y + z &= 5 \\ -5x + 7y + 2z &= -14 \end{aligned}$$

$$\begin{aligned} \text{e) } a + 10c &= 5 \\ 3a + b - 4c &= -1 \\ 4a + b + 6c &= 1 \end{aligned}$$

$$\begin{aligned} \text{f) } w - x + 2y + z &= 0 \\ 2w + 2x - z &= 0 \\ 3w + x + 2y + z &= 0 \end{aligned}$$

2) Given the following linear system:

$$\begin{aligned} x - 2y &= 1 \\ x + by &= 5 \end{aligned}$$

Determine the conditions on b such that the system has no solution.

Unit 2: Matrix Operations

2.1 – Basic Matrix Operations

A **matrix** is a rectangular array of numbers. Each number in the matrix can be known as an **entry**.

Every matrix can be described by its **size**, which is represented by $m \times n$, where m is the number rows there are in the matrix and n is the number of columns in the matrix.

For example:

matrix A has a size of 2×3

$$A = \begin{bmatrix} 3 & 1 & 6 \\ 4 & 0 & 9 \end{bmatrix}$$

Matrix B has a size of 2×2

$$B = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$

Any matrix with size $m \times m$ is known as a **square matrix**.

Each entry in a matrix has its own position. The number 3 is positioned in entry $(1,1)$ and the number 9 is positioned in entry $(2,3)$. The position is described by the row number and the column number the entry is situated in.

For example: the entries in matrix A would be labelled as follows

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{14} & a_{15} & a_{16} \end{bmatrix}$$

Two matrices are equal if they have the same size and contain identical entries.

For example: matrix A and B are equal, but A and C are not equal, and A and D are also not equal

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 6 & 8 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix}$$

Matrices can be added and multiplied in a way similar to the operations performed in the real number system. The following are basic matrix operations:

- 1) Matrix addition/ subtraction
- 2) Scalar Multiplication
- 3) Matrix Multiplication
- 4) Matrix Transposition
- 5) Trace of a square matrix

Matrix Addition/Subtraction

When adding or subtracting two matrices, their corresponding entries are added or subtracted

For example:

$$\text{Given } A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 6 & 8 \end{bmatrix} \quad A + C = \begin{bmatrix} 2 & 3 \\ 11 & 15 \end{bmatrix}$$
$$C - A = \begin{bmatrix} 0 & -3 \\ 1 & 1 \end{bmatrix}$$

$$\text{Given } C = \begin{bmatrix} 1 & 0 \\ 6 & 8 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} \quad C + D \text{ is undefined, because their sizes are different}$$

Scalar Multiplication

A number that is not in the form of a matrix is also known as a scalar.

If k is a scalar value, and A is a matrix, then kA is referred to as scalar multiplication. The scalar value, k will multiply each entry in matrix A

For example:

$$\text{Given } A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad 2A = \begin{bmatrix} 2 & 6 \\ 10 & 14 \end{bmatrix}$$
$$\text{Given } D = \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} \quad -3D = \begin{bmatrix} -15 & -6 & -3 \\ -15 & -9 & -27 \end{bmatrix}$$

Matrix Multiplication

Matrix multiplication refers to two matrices being multiplied. If A and B are matrices, then each (i, j) entry in the product matrix AB is determined by adding the products of the corresponding entries in row i of matrix A and column j of matrix B .

For example:

$$\text{Given } A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 6 & 8 \end{bmatrix}$$

The entry $(1,1)$ in product AC will be determined by adding the products of the corresponding entries in row 1 of matrix A $\begin{bmatrix} 1 & 3 \end{bmatrix}$ and column 1 of matrix B $\begin{bmatrix} 1 \\ 6 \end{bmatrix}$

Then, entry $(1, 2)$ in product AC will be determined by adding the products of the corresponding entries in row 1 of matrix $A \begin{bmatrix} 1 & 3 \end{bmatrix}$ and column 2 of matrix $B \begin{bmatrix} 0 \\ 8 \end{bmatrix}$

$$AC = \begin{bmatrix} (1)(1) + (3)(6) & (1)(0) + (3)(8) \\ (5)(1) + (7)(6) & (5)(0) + (7)(8) \end{bmatrix}$$

$$AC = \begin{bmatrix} 19 & 24 \\ 47 & 56 \end{bmatrix}$$

Matrix Transposition

Matrix transposition, or the transpose of a matrix, is derived by replacing column n by row n in a matrix.

The transpose of matrix A , which is represented by A^T , will contain a first row that is identical to the first column in A , a second row that is identical to the second column in A , and so on

For example:

$$\text{Given } A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 5 \\ 3 & 7 \end{bmatrix}$$

$$\text{Given } D = \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} \quad D^T = \begin{bmatrix} 5 & 5 \\ 2 & 3 \\ 1 & 9 \end{bmatrix}$$

If the size of a matrix is $m \times n$, then the size of its transpose would be $n \times m$

Trace of a Square Matrix

The trace of a square matrix is the sum of all the entries on the main diagonal, which is the diagonal of entries from the top left entry to bottom right entry. The trace of matrix A is represented by $tr(A)$.

For example:

$$\text{Given } A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad tr(A) = 1 + 7 = 8$$

$$\text{Given } B = \begin{bmatrix} 7 & 1 & -1 \\ 0 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \quad tr(B) = 7 + (-2) + 1 = 6$$

The result of the trace of a square matrix is a scalar value.

Example 1

Given the following matrices, complete the given operations.

$$A = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 9 \\ 1 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 7 & 3 \\ 1 & 4 & 2 \end{bmatrix} \quad D = \begin{bmatrix} 9 & 8 \\ 3 & 0 \\ 4 & -1 \end{bmatrix}$$

a) $2A + B$

$$2 \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 9 \\ 1 & 4 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 6 & 4 \\ -2 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 9 \\ 1 & 4 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 10 & 13 \\ -1 & 4 \end{bmatrix}$$

b) $A + 5D$

Undefined, because the size of A and D are not the same

c) $A \times C$

$$\begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 7 & 3 \\ 1 & 4 & 2 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} (3)(0) + (2)(1) & (3)(7) + (2)(4) & (3)(3) + (2)(2) \\ (-1)(0) + (0)(1) & (-1)(7) + (0)(4) & (-1)(3) + (0)(2) \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 2 & 29 & 13 \\ 0 & -7 & -3 \end{bmatrix}$$

d) $C \times B$

Undefined, because the number of columns in C does not equal to the number of columns in B

e) $A^T + B$

$$\begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 9 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 3 & 4 \end{bmatrix}$$

f) $tr(A)$

$$tr(A) = 3 + 0 = 3$$

g) $tr(AB)$

$$AB = \begin{bmatrix} (3)(4) + (2)(1) & (3)(9) + (2)(4) \\ (-1)(4) + (0)(1) & (-1)(9) + (0)(4) \end{bmatrix} = \begin{bmatrix} 14 & 35 \\ -4 & -9 \end{bmatrix}$$

$$tr(AB) = 14 + (-9) = 5$$

Practice

Given the following matrices, perform the given operations in each question.

$$A = \begin{bmatrix} 4 & 0 \\ -1 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 & -2 \\ -1 & 7 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 6 & 5 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & -1 \\ 4 & -9 & 9 \end{bmatrix} \quad E = \begin{bmatrix} 0 & 1 \\ 5 & 4 \\ 5 & 2 \end{bmatrix}$$

1) $2D$

2) $-3A + 5C$

3) $D - 6B$

4) $E + C$

5) $C \times A$

6) $A \times C$

7) $D \times E$

8) $C \times E$

9) D^T

10) $E^T + B$

$$11) (A+C)^T$$

$$12) \text{tr}(A)$$

$$13) \text{tr}(AC)$$

$$14) A^2 \text{ (hint: } A^2 = A \times A \text{)}$$

2.2 - Algebraic Properties of Matrices

The following are algebraic properties of matrices:

Given A , B , and C are matrices, and k and p are a scalar

Matrix Addition

$$A+B=B+A$$

Commutative property

$$A+(B+C)=(A+B)+C$$

Associative property

$$A+0=A$$

Where 0 represents the zero matrix (all entries are zero)

$$A+(-B)=A-B$$

$$A-A=0$$

Scalar Multiplication

$$k(A+B)=kA+kB$$

Distributive property

$$(k+p)A=kA+pA$$

Distributive property

$$(kp)A=k(pA)$$

Associative property

$$k(AB)=kA(B)=A(kB)$$

Associative property

$$1A=A$$

Matrix Multiplication

$$AB \neq BA$$

The commutative property does not hold for matrix multiplication

$$A(BC) = (AB)C$$

Associative property

$$AI = IA = A$$

Where I is known as the Identity Matrix

$$A(B+C) = AB + AC$$

Distributive property

Matrix Transposition

$$(A^T)^T = A$$

$$(kA)^T = k A^T$$

$$(A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

Trace

$$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\text{tr}(kA) = k \text{tr}(A)$$

$$\text{tr}(A^T) = \text{tr}(A)$$

$$\text{tr}(AB) = \text{tr}(BA)$$

The Zero and Identity Matrices

The Zero Matrix is a matrix consisting of zeroes in all its entries.

For example, the following are all zero matrices

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The Identity Matrix is a square matrix of size $n \times n$ that consists of 1's on its main diagonal and 0's in all the other entries.

For example, the following are Identity matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 1

Using the given matrices, show the commutative property of matrix addition

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 5 \\ 5 & 0 \end{bmatrix}$$

The commutative property of matrix addition states that $A + B = B + A$

$$A + B = \begin{bmatrix} 5 & 8 \\ 10 & 7 \end{bmatrix}$$

$$B + A = \begin{bmatrix} 5 & 8 \\ 10 & 7 \end{bmatrix}$$

$$A + B = B + A$$

Example 2

Using the given matrices and scalar value, show the distributive property of scalar multiplication

$$D = \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} \quad E = \begin{bmatrix} -1 & 2 & 0 \\ 12 & 1 & 3 \end{bmatrix} \quad k = 2$$

The distributive property of scalar multiplication states that $k(D + E) = kD + kE$

$$k(D + E) = 2 \left(\begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 0 \\ 12 & 1 & 3 \end{bmatrix} \right) = 2 \left(\begin{bmatrix} 4 & 4 & 1 \\ 17 & 4 & 12 \end{bmatrix} \right) = \begin{bmatrix} 8 & 8 & 2 \\ 34 & 8 & 24 \end{bmatrix}$$

$$kD + kE = 2 \begin{bmatrix} 5 & 2 & 1 \\ 5 & 3 & 9 \end{bmatrix} + 2 \begin{bmatrix} -1 & 2 & 0 \\ 12 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 4 & 2 \\ 10 & 6 & 18 \end{bmatrix} + \begin{bmatrix} -2 & 4 & 0 \\ 24 & 2 & 6 \end{bmatrix} = \begin{bmatrix} 8 & 8 & 2 \\ 34 & 8 & 24 \end{bmatrix}$$

$$k(D + E) = kD + kE$$

Example 3

Using the given matrix, show that $AI = A$

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$

$$AI = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$

Practice

Given the following matrices, show that the following algebraic properties hold true.

$$A = \begin{bmatrix} 4 & 0 \\ -1 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 6 & 5 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 10 & 6 \end{bmatrix}$$

- 1) $A + (-B) = A - B$
- 2) The associative property of matrix multiplication
- 3) $AB \neq BA$
- 4) $BI = IB = B$
- 5) $(A^T)^T = A$
- 6) $0 + C = C$
- 7) $\text{tr}(C + B) = \text{tr}(C) + \text{tr}(B)$
- 8) $(A + B)^T = A^T + B^T$
- 9) $(k + p)A = kA + pA$ for $k = -1$ and $p = 4$
- 10) $(AB)^T = B^T A^T$

Unit 3: Inverse of a Matrix

3.1 - Introduction to Inverse of a Matrix

The **inverse** of a square matrix A , represented by A^{-1} , is a matrix such that $AA^{-1} = I$

In other words, when a matrix is multiplied by its inverse, the product is the Identity matrix.

So if there are two square matrices, A and B , and the following is true:

$$AB=I \text{ and } BA=I$$

Then, A and B are inverses of one another

When a matrix has an inverse, it is known as an **invertible matrix**.

Example 1

Show that the following matrices are inverses of one another.

$$A=\begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \quad B=\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

$$AB=\begin{bmatrix} (2)(3)+(-5)(1) & (2)(5)+(-5)(2) \\ (-1)(3)+(3)(1) & (-1)(5)+(3)(2) \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA=\begin{bmatrix} (3)(2)+(5)(-1) & (3)(-5)+(5)(3) \\ (1)(2)+(2)(-1) & (1)(-5)+(2)(3) \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AB=BA=I$$

Therefore, A and B are inverses of one another.

Invertible 2x2 Matrices

If a 2x2 matrix has the following entries $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, it is invertible if $ad - bc \neq 0$. In other words, if the value of $ad - bc$ is zero, the matrix will not have an inverse.

Example 2

Determine whether the following 2x2 matrix is invertible.

$$\begin{bmatrix} 1 & -2 \\ 5 & -1 \end{bmatrix}$$

$$ad - bc = (1)(-1) - (-2)(5) = -1 + 10 = 9$$

Since $ad - bc \neq 0$, the matrix is invertible.

Practice

1) Determine whether the following are inverses of one another.

a) $A = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

b) $A = \begin{bmatrix} 1 & 7 \\ -2 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & \frac{-1}{2} \\ \frac{1}{7} & \frac{1}{14} \end{bmatrix}$

c) $A = \begin{bmatrix} 9 & -1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 2 \\ 1 & -2 \end{bmatrix}$

d) $A = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 2 \\ 1 & -2 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 4 & -2 & -5 \\ 2 & -1 & -3 \\ 1 & 0 & -1 \end{bmatrix}$

e) $A = \begin{bmatrix} 2 & -2 & -1 \\ 1 & 2 & 0 \\ 9 & -3 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0.5 & -1 \\ 6 & -1 & 1 \\ 7 & 1 & 0 \end{bmatrix}$

2) Determine whether the following 2x2 matrices are invertible

a) $\begin{bmatrix} 3 & 6 \\ 2 & 4 \end{bmatrix}$

b) $\begin{bmatrix} 9 & -1 \\ 8 & 1 \end{bmatrix}$

c) $\begin{bmatrix} -5 & 2 \\ 3 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 3 & 0 \\ 2 & 0 \end{bmatrix}$

3.2 – Inverse of a 2x2 Matrix

In the previous chapter, we learned to determine whether a 2x2 matrix is invertible. If matrix A is invertible, it would mean that there exists a matrix A^{-1} of which when multiplied, the product is the Identity matrix I . In this chapter, we will learn to determine the inverse matrix, A^{-1} .

Given a 2x2 matrix, A , where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse matrix, A^{-1} , can be determine using the following formula:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

In the last chapter, it was discussed that A is an invertible 2x2 matrix when $ad-bc \neq 0$.

Observe the formula, if $ad-bc=0$, then $\frac{1}{ad-bc}$ would be undefined. Therefore, A^{-1} does not exist, meaning that A is not invertible.

Example 1

Determine the inverse of the given matrix.

$$B = \begin{bmatrix} 5 & -1 \\ -4 & 3 \end{bmatrix}$$

$$B^{-1} = \frac{1}{(5)(3) - (-1)(-4)} \begin{bmatrix} 3 & 1 \\ 4 & 5 \end{bmatrix}$$

$$B^{-1} = \frac{1}{15-4} \begin{bmatrix} 3 & 1 \\ 4 & 5 \end{bmatrix}$$

$$B^{-1} = \frac{1}{11} \begin{bmatrix} 3 & 1 \\ 4 & 5 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} \frac{3}{11} & \frac{1}{11} \\ \frac{4}{11} & \frac{5}{11} \end{bmatrix}$$

Example 2

Determine A^{-2} given $A = \begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix}$

Rewriting using the inverse $A^{-2} = A^{-1} A^{-1}$

Finding the inverse

$$A^{-1} = \frac{1}{(2)(1) - (-3)(1)} \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{(2)(1) - (-3)(1)} \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

Perform multiplication

$$A^{-1} A^{-1} = \left(\frac{1}{5} \right) \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \times \left(\frac{1}{5} \right) \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

$$A^{-1} A^{-1} = \left(\frac{1}{5} \right) \left(\frac{1}{5} \right) \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

$$A^{-1} A^{-1} = \left(\frac{1}{25} \right) \begin{bmatrix} (1)(1) + (-1)(3) & (1)(-1) + (-1)(2) \\ (3)(1) + (2)(3) & (3)(-1) + (2)(2) \end{bmatrix}$$

$$A^{-1} A^{-1} = \left(\frac{1}{25} \right) \begin{bmatrix} -2 & -3 \\ 9 & 1 \end{bmatrix}$$

$$A^{-2} = \begin{bmatrix} \frac{-2}{25} & \frac{-3}{25} \\ \frac{9}{25} & \frac{1}{25} \end{bmatrix}$$

Practice

1) Given $A = \begin{bmatrix} 3 & 0 \\ 1 & -4 \end{bmatrix}$, determine A^{-1}

2) Given $B = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$, determine B^{-1}

3) Given $C = \begin{bmatrix} 1 & 4 \\ -1 & -4 \end{bmatrix}$, determine C^{-1}

4) Given $A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$, determine A^{-2}

5) Given $H = \begin{bmatrix} 8 & 3 \\ 1 & -1 \end{bmatrix}$, determine $(H^{-1})^T$

6) Given $A = \begin{bmatrix} 1 & 9 \\ 1 & 8 \end{bmatrix}$, determine $(A^T)^{-1}$

7) Given $A = \begin{bmatrix} 1 & 9 \\ 1 & 8 \end{bmatrix}$, determine $(A^{-1})^{-1}$

8) Given $B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$, determine B^{-1}

9) Determine the inverse of the 2x2 Identity matrix.

10) Is the 2x2 zero matrix invertible? Explain why or why not.

3.3 - Inverse of a 3x3 Matrix

Unlike the procedure taken to determine the inverse of a 2x2 matrix, finding the inverse of a 3x3 matrix requires performing elementary row operations.

Recall from Chapter 1.1, elementary row operations are the following:

- 1) Multiplying an entire row by a non-zero constant
- 2) Interchanging two rows
- 3) Adding or subtracting two rows

To determine the inverse of a 3x3 matrix, we will make use of the **Matrix Inversion Algorithm**:

If a square matrix A is invertible, then there exists a sequence of elementary row operations that transforms matrix A to the Identity matrix I of the same size. That series of elementary row operations will transform the Identity matrix I to A^{-1} when the elementary row operations are carried out simultaneously.

Example 1

Determine the inverse of the given matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -5 & 7 & -11 \\ -2 & 3 & -5 \end{bmatrix}$$

First write A alongside with I

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ -5 & 7 & -11 & 0 & 1 & 0 \\ -2 & 3 & -5 & 0 & 0 & 1 \end{array} \right]$$

Elementary row operations will be performed to transform A into I

Once completed, the matrix on the left will be I and the matrix on the right will be A^{-1}

$$R_2' = 5R_1 + R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 5 & 1 & 0 \\ -2 & 3 & -5 & 0 & 0 & 1 \end{array} \right]$$

$$R_3' = 2R_1 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 5 & 1 & 0 \\ 0 & 1 & 1 & 2 & 0 & 1 \end{array} \right]$$

Interchange R_2 and R_3

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 1 \\ 0 & 2 & -1 & 5 & 1 & 0 \end{array} \right]$$

$$R_1' = R_1 + R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 3 & 0 & 1 \\ 0 & 1 & 1 & 2 & 0 & 1 \\ 0 & 2 & -1 & 5 & 1 & 0 \end{array} \right]$$

$$R_3' = 2R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 3 & 0 & 1 \\ 0 & 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & -1 & -1 & 2 \end{array} \right]$$

$$R_1' = R_1 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 1 & -1 \\ 0 & 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & -1 & -1 & 2 \end{array} \right]$$

$$R_3' = R_3 \div 3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 1 & -1 \\ 0 & 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & \frac{-1}{3} & \frac{-1}{3} & \frac{2}{3} \end{array} \right]$$

$$R_2' = R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} & & & 4 & 1 & -1 \\ 1 & 0 & 0 & \frac{7}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 & \frac{7}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 & \frac{-1}{3} & \frac{-1}{3} & \frac{2}{3} \end{array} \right]$$

The matrix on the left is now I , thus, the matrix on the right is the inverse of A

$$A^{-1} = \begin{bmatrix} 4 & 1 & -1 \\ \frac{7}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{-1}{3} & \frac{-1}{3} & \frac{2}{3} \end{bmatrix}$$

Example 2

Determine the inverse of the given matrix

$$B = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

First write B alongside with I

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$R_2' = 2R_1 - R_2$$

$$R_3' = R_1 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 8 & 9 & 2 & -1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right]$$

$$R_2' = R_2 \div 8$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 1 & \frac{9}{8} & \frac{1}{4} & \frac{-1}{8} & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right]$$

$$R_3' = 8R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 1 & \frac{9}{8} & \frac{1}{4} & \frac{-1}{8} & 0 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{array} \right]$$

Because R_3 became zeros, there isn't a way to generate a 1 in the entry (3,3). Thus, the matrix on the left cannot be transformed into the Identity matrix. This means that matrix B does not have an inverse, or in other words, not invertible.

Practice

Given the following 3x3 matrices, determine the inverse.

$$1) \begin{bmatrix} 1 & 0 & 0 \\ 7 & 0 & 2 \\ 3 & 5 & 1 \end{bmatrix}$$

$$2) \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 0 \\ 1 & 5 & -1 \end{bmatrix}$$

$$3) \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$4) \begin{bmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{bmatrix}$$

$$5) \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$$6) \begin{bmatrix} 3 & 1 & -1 \\ 5 & 2 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

$$7) \begin{bmatrix} 2 & 4 & 1 \\ 3 & 3 & 2 \\ 4 & 1 & 4 \end{bmatrix}$$

3.4 – Solving a Linear System using the Inverse of a Matrix

Linear systems can be solved using the inverse of a matrix by first writing it as an equation involving matrices: $AX=C$

For example

Consider the following linear system:

$$2x - y = 7$$

$$5x - 3y = 18$$

This linear system could be represented using matrices as follows:

$$\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

because the product of

$$\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \text{ is } \begin{bmatrix} 2x - y \\ 5x - 3y \end{bmatrix}$$

To solve the above linear system, $\begin{bmatrix} x \\ y \end{bmatrix}$ needs to be solved.

In the equation $\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$, let $A = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

To solve for $\begin{bmatrix} x \\ y \end{bmatrix}$, we apply the inverse of A

$$A^{-1} A \begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

$$I \begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

The next step is to determine A^{-1} and find the product of $A^{-1} \begin{bmatrix} 7 \\ 18 \end{bmatrix}$

Since A is a 2x2 matrix, the formula $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ is used

$$A^{-1} = \frac{1}{(2)(-3) - (5)(-1)} \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$$

$$A^{-1} = -1 \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ 5 & -2 \end{bmatrix}$$

So

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 21 - 18 \\ 35 - 36 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Therefore, $x=3$ and $y=-1$

In summary, the following steps are taken to solve a linear system using the inverse of a matrix:

- 1) Write the matrix of coefficients, A , using the coefficients of the linear equations
- 2) Write the matrix of constants, C , using the constants of the linear equations
- 3) Write the matrix, X , which contains the unknowns of the linear system
- 4) Write the linear system in the form of $AX=C$
- 5) Multiply both sides by the inverse A^{-1} to obtain $A^{-1}AX=A^{-1}C$
- 6) Isolate for X by using $A^{-1}A=I$, obtaining $X=A^{-1}C$
- 7) Determine A^{-1}
- 8) Solve for X by multiplying $A^{-1}C$

Note: if matrix A is not invertible, then the linear system either has no solution or infinitely many solutions.

Example 1

Solve the given linear system using the inverse of a matrix

$$\begin{aligned} -2x + y + 3z &= 1 \\ 3x - y + z &= 5 \\ -5x + 2z &= -1 \end{aligned}$$

In the form of $AX=C$, the linear system is written as follows:

$$\begin{bmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

$$A^{-1} \begin{bmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

Since A is a 3x3 matrix, the matrix inversion algorithm is used to find

$$A^{-1} = \begin{bmatrix} \frac{-1}{11} & \frac{-1}{11} & \frac{-2}{11} \\ \frac{1}{2} & \frac{-1}{2} & \frac{-1}{2} \\ \frac{5}{22} & \frac{5}{22} & \frac{1}{22} \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{-4}{11} \\ \frac{-3}{2} \\ \frac{29}{22} \end{bmatrix}$$

The solution is $x = \frac{-4}{11}$, $y = \frac{-3}{2}$, $z = \frac{29}{22}$

Practice

Solve the given linear systems using the inverse of a matrix

1) $x + 2y = 3$
 $3x + 4y = 11$

2) $5x + 3y = 52$

$$2x - 4y = -26$$

$$\begin{aligned} 3) \quad & x + y = 5 \\ & 2x - 3y = -5 \end{aligned}$$

$$\begin{aligned} 4) \quad & 5x + 6y = 13 \\ & 15x + 18y = 39 \end{aligned}$$

$$\begin{aligned} 5) \quad & 10x - 2y = -3 \\ & y - 5x = 1 \end{aligned}$$

$$\begin{aligned} 6) \quad & a + 10c = 5 \\ & 3a + b - 4c = -1 \\ & 4a + b + 6c = 1 \end{aligned}$$

$$\begin{aligned} 7) \quad & 4x + y = -4 \\ & 4x - y + 3z = 5 \\ & x + y + z = 3 \end{aligned}$$

$$\begin{aligned} 8) \quad & x + y + z = -6 \\ & 3x + 4y = 1 \\ & 2y = 5z \end{aligned}$$

Unit 4: Special Matrices and Determinants

4.1 – Special Matrices and Operations

There are matrices that have a unique form and unique properties when taken operations upon.

In this chapter, Special Matrices to be discussed will include:

- 1) Square Matrices
- 2) Symmetric Matrices
- 3) Diagonal Matrices
- 4) Triangular Matrices

Square Matrices

Square matrices are matrices that have size $n \times n$

For example, the Identity Matrix is always a square matrix.

Any matrix with size 2x2, 3x3, 4x4, and so on... are all square matrices

Symmetric Matrices

A Symmetric Matrix is a square matrix A such that $A = A^T$, where the matrix itself is equal to its transpose.

For example, the Identity Matrix is a symmetric matrix because $I = I^T$

For example, this matrix, $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$, is also a symmetric matrix because it is the same as its transpose.

Diagonal Matrices

A Diagonal Matrix is a square matrix such that entries off the main diagonal are zero. Recall that the main diagonal is the diagonal from the top left to the bottom right.

For example, the Identity Matrix is a diagonal matrix. It contains 1's on the main diagonal, and entries off the main diagonal are zero.

For example, this matrix, $A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ has zeros on all entries off the diagonal

$B = \begin{bmatrix} 7 & 0 \\ 0 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are also Diagonal Matrices.

Triangular Matrices

A Triangular Matrix is a square matrix such that entries below or above the main diagonal are zeros.

For example: $A = \begin{bmatrix} 7 & 0 \\ 5 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 7 & 9 \\ 0 & -1 \end{bmatrix}$ $C = \begin{bmatrix} 3 & 1 & 9 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 2 & 5 & 1 \end{bmatrix}$

Triangular matrices could either be an **Upper Triangular Matrices** or a **Lower Triangular Matrices**.

An upper triangular matrix is a triangular matrix with all entries below the main diagonal as zeros. From the above, B and C are upper triangular matrices

A lower triangular matrix is a triangular matrix with all entries above the main diagonal as zeros. From the above, A and D are lower triangular matrices

Example 1

Identify the following special matrices.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 10 & 1 \\ 0 & 0 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 7 & 0 \\ 0 & 9 \end{bmatrix}$$

A is a diagonal matrix

B is an upper triangular matrix

C is a symmetric matrix

D is a diagonal matrix

Properties of Special Matrices

Inverses

- 1) A diagonal matrix is invertible if and only if all of its entries on the main diagonal are non-zero values
- 2) A triangular matrix is invertible if and only if all of its entries on the main diagonal are non-zero values

$$3) \text{ The Inverse of a diagonal matrix, } A = \begin{bmatrix} a & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k \end{bmatrix}, A^{-1} = \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k^{-1} \end{bmatrix}$$

- 4) The inverse of an invertible lower triangular matrix is a lower triangular matrix
- 5) The inverse of an invertible upper triangular matrix is an upper triangular matrix

Sums

- 6) If A and B are symmetric and are the same size, then the sum $A+B$ or $A-B$ are also symmetric
- 7) If A is symmetric and k is a scalar value, then kA is symmetric

Products

$$8) \text{ If a matrix } A \text{ is a diagonal matrix, } A = \begin{bmatrix} a & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k \end{bmatrix}, \text{ then } A^n = \begin{bmatrix} a^n & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k^n \end{bmatrix}$$

- 9) The product of two lower triangular matrices is a lower triangular matrix
- 10) The product of two upper triangular matrices is an upper triangular matrix.

Transpose

11) The transpose of an upper triangular matrix is a lower triangular matrix, and vice versa

Example 2

Given, the matrix $D = \begin{bmatrix} 7 & 0 \\ 0 & 9 \end{bmatrix}$, determine D^{-1} using property 3

Because matrix D is a diagonal matrix, $D^{-1} = \begin{bmatrix} 7^{-1} & 0 \\ 0 & 9^{-1} \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & 0 \\ 0 & \frac{1}{9} \end{bmatrix}$

Example 3

Given the matrix $C = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, show that kC is also symmetric.

Matrix C is a symmetric matrix because $C = C^T = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$kC = \begin{bmatrix} 3k & k & k \\ k & 2k & k \\ k & k & 0 \end{bmatrix}$, which is a symmetric matrix because $(kC)^T = kC = \begin{bmatrix} 3k & k & k \\ k & 2k & k \\ k & k & 0 \end{bmatrix}$

Practice

1) Given the following matrices

$$\begin{aligned} A &= \begin{bmatrix} 0 & 0 & 2 \\ 9 & 4 & 1 \end{bmatrix} & B &= \begin{bmatrix} 9 & 1 & 5 \\ 1 & 7 & 4 \\ 5 & 4 & 0 \end{bmatrix} & C &= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} & D &= \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} & E &= \begin{bmatrix} 8 & 4 & 5 \\ 0 & 9 & 2 \\ 0 & 0 & -1 \end{bmatrix} \\ F &= \begin{bmatrix} 8 & 0 & 0 \\ 7 & 1 & 0 \\ 1 & -5 & 7 \end{bmatrix} & G &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \\ 6 & 11 & 6 \end{bmatrix} & H &= \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -6 \end{bmatrix} & J &= \begin{bmatrix} 12 & 0 & 0 \\ 0 & 36 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

- a) State the matrices that are square matrices
 - b) State the matrices that are symmetric matrices
 - c) State the matrices that are diagonal matrices
 - d) State the matrices that are triangular matrices
 - e) State the matrices that are upper triangular matrices
 - f) State the matrices that are lower triangular matrices
-
- 2) Determine the inverse of matrix H using property 3
 - 3) Determine J^2 using property 8
 - 4) Determine the inverse of matrix F and show that F^{-1} is also a lower triangular matrix
 - 5) Demonstrate property 6 by adding $B+H$
 - 6) Demonstrate property 11 using matrix E
 - 7) State whether the following statements are true or false
 - a) All triangular matrices are square matrices
 - b) Some square matrices are diagonal matrices
 - c) Some diagonal matrices are triangular matrices
 - d) All triangular matrices are upper triangular matrices

4.2 - The Adjoint Theorem

The determinant is one of the concepts that is used in the applications of linear transformations.

To calculate the determinant of a matrix, the calculations of a matrix known as the **Adjoint** needs to be calculated first.

The Adjoint of a square matrix can be determined using the following theorem:

$$Adj(A) = [c_{ij}(A)]^T$$

where $Adj(A)$ represents the adjoint of a matrix A

and c_{ij} represents the cofactors of each entry of matrix A

and $c_{ij}(A)$ represents the cofactor matrix of matrix A

Thus, the adjoint is determined by the transpose of the cofactor matrix.

Recall in Chapter 3.2, the inverse of a 2×2 matrix A can be found using the following formula:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

This formula could be rewritten as

$$A^{-1} = \frac{1}{\det(A)} (\text{Adj}(A))$$

because in actual fact, the determinant is $\det(A) = ad - bc$ and the adjoint is $\text{Adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

when A is a 2×2 matrix.

However, calculations become more complicated for square matrices that are larger than 2×2 and the adjoint needs to be determined using the above theorem.

Calculating Cofactors

The cofactor of an entry is determined using the following formula:

$$c_{ij} = (-1)^{i+j} m_{ij}$$

where

i is the row the entry is located in

j is the column the entry is located in

m_{ij} is the minor of the entry, which is calculated using $ad - bc$ of the 2×2 matrix created after deleting row i and column j

For example, given a matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & 5 \\ 0 & 7 & 2 \end{bmatrix}$, the cofactor of entry a_{13} would be calculated by

$$c_{13} = (-1)^{1+3} m_{13}$$

m_{13} would be obtained from the calculation of $ad - bc$ in the 2×2 matrix created after deleting row 1 and column 3

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & 5 \\ 0 & 7 & 2 \end{bmatrix}$$

$$m_{13} = (2)(7) - (0)(3) = 14$$

So

$$c_{13}=(-1)^5(14)=-14$$

Calculating the Adjoint

Given a 3×3 matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, the adjoint would be $Adj(A) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}^T$

Example

Calculate the adjoint of $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & 5 \\ 0 & 7 & 2 \end{bmatrix}$

Cofactors will be calculated first

$$c_{11}=(-1)^{1+1}[(3)(2)-(7)(5)]=-29$$

$$c_{12}=(-1)^{1+2}[(2)(2)-(0)(5)]=-4$$

$$c_{13}=(-1)^{1+3}[(2)(7)-(0)(3)]=14$$

$$c_{21}=(-1)^{2+1}[(0)(2)-(7)(3)]=21$$

$$c_{22}=(-1)^{2+2}[(1)(2)-(0)(3)]=2$$

$$c_{23}=(-1)^{2+3}[(1)(7)-(0)(0)]=-7$$

$$c_{31}=(-1)^{3+1}[(0)(5)-(3)(3)]=-9$$

$$c_{32}=(-1)^{3+2}[(1)(5)-(2)(3)]=1$$

$$c_{33}=(-1)^{3+3}[(1)(3)-(2)(0)]=3$$

So the adjoint is $Adj(A) = \begin{bmatrix} -29 & -4 & 14 \\ 21 & 2 & -7 \\ -9 & 1 & 3 \end{bmatrix}^T = \begin{bmatrix} -29 & 21 & -9 \\ -4 & 2 & 1 \\ 14 & -7 & 3 \end{bmatrix}$

Practice

1) Given the following matrix, determine the following cofactors

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 0 & 6 & 9 \\ -1 & 2 & 3 \end{bmatrix}$$

a) c_{32}

b) c_{21}

c) c_{11}

d) c_{23}

2) Given the following matrices, determine the adjoint

a) $A = \begin{bmatrix} 6 & 5 \\ 0 & 1 \end{bmatrix}$

b) $B = \begin{bmatrix} -1 & 2 \\ -8 & 1 \end{bmatrix}$

c) $C = \begin{bmatrix} 8 & 0 & 1 \\ 1 & -1 & 2 \\ 0 & 5 & 1 \end{bmatrix}$

d) $D = \begin{bmatrix} 3 & 1 & 5 \\ 0 & 11 & 4 \\ 9 & -1 & 1 \end{bmatrix}$

e) $E = \begin{bmatrix} 0 & 7 & 2 \\ 8 & 1 & 5 \\ 1 & 6 & 5 \end{bmatrix}$

4.3 - The Determinant

The determinant was briefly discussed in previous chapter. It is useful in helping us find the inverse of a matrix and also in solving linear systems.

Recall, given a 2×2 matrix A , $\det(A) = ad - bc$

For a 3×3 matrix, calculating the determinant will require using a method known as **Cofactor Expansion**.

In this method, the Cofactor Expansion formula is applied along any selected row or column of the matrix. Multiply each entry of that selected row or column by the corresponding cofactor and then determine the sum.

For example, given a matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, if row 1 is selected then performing cofactor expansion along row 1 would mean following the formula:

$$\det(A) = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

If cofactor expansion is performed along row 2, then the following formula applies:

$$\det(A) = a_{21}c_{21} + a_{22}c_{22} + a_{23}c_{23}$$

If cofactor expansion is performed along column 3, then the following formula applies:

$$\det(A) = a_{13}c_{13} + a_{23}c_{23} + a_{33}c_{33}$$

Example 1

Given the following matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 5 \\ 3 & 7 & 2 \end{bmatrix}$, calculate the determinant by performing cofactor expansion along row 1.

Since the cofactor expansion will be performed along row 1, the following formula will be applied

$$\det(A) = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

$$\det(A) = (1)c_{11} + (1)c_{12} + (3)c_{13}$$

The calculation of the corresponding cofactors is necessary.

$$c_{11} = (-1)^{1+1}[(3)(2) - (7)(5)] = -29$$

$$c_{12} = (-1)^{1+2}[(2)(2) - (3)(5)] = 11$$

$$c_{13}=(-1)^{1+3}[(2)(7)-(3)(3)]=5$$

So substituting the cofactors into the formula: $\det(A)=(1)(-29)+(1)(11)+(3)(5)$

$$\det(A)=-29+11+15$$

$$\det(A)=-3$$

The determinant of matrix A is -3 .

Example 2

Given the following matrix $A=\begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 5 \\ 3 & 7 & 2 \end{bmatrix}$, calculate the determinant by performing cofactor expansion along column 2.

Since the cofactor expansion will be performed along column 2, the following formula will be applied

$$\det(A)=a_{12}c_{12}+a_{22}c_{22}+a_{32}c_{32}$$

$$\det(A)=(1)c_{12}+(3)c_{22}+(7)c_{32}$$

The calculation of the corresponding cofactors is necessary.

$$c_{12}=(-1)^{1+2}[(2)(2)-(3)(5)]=11$$

$$c_{22}=(-1)^{2+2}[(1)(2)-(3)(3)]=-7$$

$$c_{32}=(-1)^{2+3}[(1)(7)-(3)(1)]=-4$$

The determinant is

$$\det(A)=(1)(11)+(3)(-7)+(7)(-4)=-38$$

Example 3

Given the following matrix $A=\begin{bmatrix} 1 & 0 & 30 \\ 0 & 3 & 0 \\ -3 & 17 & 21 \end{bmatrix}$, calculate the determinant by performing cofactor expansion.

In this case, any row or column can be selected to perform the cofactor expansion. Notice that row 2 contains two entries of zeros. For convenience, row 2 will be selected.

$$\det(A) = a_{21}c_{21} + a_{22}c_{22} + a_{23}c_{23}$$

$$\det(A) = (0)c_{21} + (3)c_{22} + (0)c_{23}$$

Since the product of zero and any other number is zero, in this case, only the cofactor c_{22} needs to be calculated.

$$c_{22} = (-1)^{2+2}[(1)(21) - (-3)(30)] = 111$$

The determinant is

$$\det(A) = 0 + (3)(111) + 0 = 333$$

Practice

- 1) Calculate the determinant of the following matrices

a) $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

b) $A = \begin{bmatrix} 4 & -5 \\ 0 & 12 \end{bmatrix}$

- 2) Given the following matrix, calculate the determinant by expanding along the following:

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 0 & 6 & 9 \\ -1 & 2 & 3 \end{bmatrix}$$

- a) Row 3
- b) Column 2
- c) Row 1
- d) Column 3

- 3) Given the following matrices, calculate the determinant using cofactor expansion.

a) $A = \begin{bmatrix} 2 & 0 & 6 \\ 1 & -1 & 0 \\ 1 & -5 & 1 \end{bmatrix}$

$$\text{b) } B = \begin{bmatrix} 4 & 1 & 1 \\ 0 & 8 & 1 \\ 9 & 0 & 7 \end{bmatrix}$$

$$\text{c) } C = \begin{bmatrix} 0 & 7 & -2 \\ 3 & 1 & 5 \\ 1 & 8 & -5 \end{bmatrix}$$

4.4 – Cramer's Rule

In Chapter 3.4, we have seen the way the inverse of a matrix is useful in solving a linear system, $AX=C$. Another method to solve a linear system that involves the inverse of a matrix is known as

Cramer's Rule:

If $AX=C$ is a system of n linear equations in n unknowns such that $\det(A) \neq 0$, then the system has a unique solution. This solution is

$$x_1 = \frac{\det(A_1)}{\det(A)}, x_2 = \frac{\det(A_2)}{\det(A)}, \dots, x_n = \frac{\det(A_n)}{\det(A)}$$

where A_j is the matrix obtained by replacing the entries in the j^{th} column of A by the entries in the matrix of constants

$$C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Example 1

Solve the following linear system using Cramer's Rule

$$2x - y = 7$$

$$5x - 3y = 18$$

This linear system could be represented using matrices as follows:

$$\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

where $A = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$ and $C = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$.

According to Cramer's Rule:

$$x_1 = \frac{\det(A_1)}{\det(A)}, x_2 = \frac{\det(A_2)}{\det(A)}$$

$$x_1 = \frac{\det\left(\begin{bmatrix} 7 & -1 \\ 18 & -3 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\right)}, x_2 = \frac{\det\left(\begin{bmatrix} 2 & 7 \\ 5 & 18 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\right)}$$

where A_1 is a matrix obtained from replacing column 1 with C and A_2 is a matrix obtained from replacing column 2 with C .

Calculating the determinants will result in the following solution:

$$x_1 = \frac{\det\left(\begin{bmatrix} 7 & -1 \\ 18 & -3 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\right)} = \frac{-21 - (-18)}{-6 - (-5)} = \frac{-3}{-1} = 3$$

$$x_2 = \frac{\det\left(\begin{bmatrix} 2 & 7 \\ 5 & 18 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}\right)} = \frac{36 - 35}{-6 - (-5)} = \frac{1}{-1} = -1$$

The solutions are $x=3$ and $y=-1$

Example 2

Solve the following linear system using Cramer's Rule

$$\begin{aligned} -2x + y + 3z &= 1 \\ 3x - y + z &= 5 \\ -5x + 2z &= -1 \end{aligned}$$

In the form of $AX = C$, the linear system is written as follows:

$$\begin{bmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

According to Cramer's Rule

$$x_1 = \frac{\det(A_1)}{\det(A)}, x_2 = \frac{\det(A_2)}{\det(A)}, x_3 = \frac{\det(A_3)}{\det(A)}$$

$$x_1 = \frac{\det\begin{pmatrix} 1 & 1 & 3 \\ 5 & -1 & 1 \\ -1 & 0 & 2 \end{pmatrix}}{\det\begin{pmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{pmatrix}}, x_2 = \frac{\det\begin{pmatrix} -2 & 1 & 3 \\ 3 & 5 & 1 \\ -5 & -1 & 2 \end{pmatrix}}{\det\begin{pmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{pmatrix}}, x_3 = \frac{\det\begin{pmatrix} -2 & 1 & 1 \\ 3 & -1 & 5 \\ -5 & 0 & -1 \end{pmatrix}}{\det\begin{pmatrix} -2 & 1 & 3 \\ 3 & -1 & 1 \\ -5 & 0 & 2 \end{pmatrix}}$$

For each determinant of the 3×3 matrix, cofactor expansion is needed. In most of the matrices, row 3 contains an entry of zero, cofactor expansion will be performed along row 3.

$$\det(A) = (-5)(-1)^{3+1}(1+3) + (0) + (2)(-1)^{3+3}(2-3) = -22$$

$$\det(A_1) = (-1)(-1)^{3+1}(1+3) + (0) + (2)(-1)^{3+3}(-1-5) = 8$$

$$\det(A_2) = (-5)(-1)^{3+1}(1-15) + (-1)(-1)^{3+2}(-2-9) + (2)(-1)^{3+3}(-10-3) = 33$$

$$\det(A_3) = (-5)(-1)^{3+1}(5+1) + (0) + (-1)(-1)^{3+3}(2-3) = -29$$

So

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{8}{-22} = -\frac{4}{11}$$

$$x_2 = \frac{\det(A_2)}{\det(A)} = \frac{33}{-22} = -\frac{3}{2}$$

$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{-29}{-22} = \frac{29}{22}$$

The solution is $x = -\frac{4}{11}$, $y = -\frac{3}{2}$, $z = \frac{29}{22}$

Practice

Using Cramer's Rule, solve the following linear systems

- 1) $x+2y=3$
 $3x+4y=11$
- 2) $5x+3y=52$
 $2x-4y=-26$
- 3) $x+y=5$
 $2x-3y=-5$
- 4) $a+10c=5$
 $3a+b-4c=-1$
 $4a+b+6c=1$
- 5) $4x+y=-4$
 $4x-y+3z=5$
 $x+y+z=3$
- 6) $x+y+z=-6$
 $3x+4y=1$
 $2y=5z$

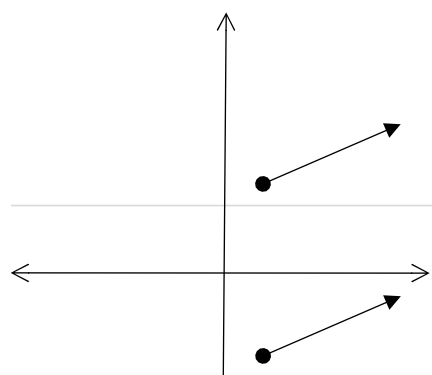
Unit 5: Vectors in 2, 3, and n-Space

5.1 - Performing Operations using Vector Components

A **vector** is formed when a line segment between two points, A and B , has a direction also (from point A to B). Quantities that are specified by both a magnitude and a direction are known as **vector quantities**. Examples of vector quantities are displacement, velocity, and force. Velocity is a vector quantity because the speed at which an object is moving would be the magnitude of the velocity vector and it is also described by the direction at which the object is moving. A quantity that is not a vector quantity is known as a scalar quantity.

Geometrically, a vector in a 2-dimensional space is depicted in the following diagram.

This vector is directed from point A to B and will be named as $\overrightarrow{AB}$.

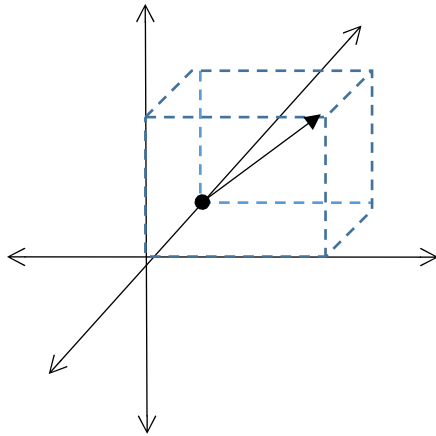


Point A is known as the tail of the vector and point B is known as the tip. The magnitude of the vector, represented by $|\overrightarrow{AB}|$ is the length of the vector. Two vectors are equal if and only if they have the same magnitude and direction. Geometrically,

two vectors can be the same even if their tips and tails are different. In the above diagram, vectors $\overrightarrow{AB}$ and $\vec{u}$ are equal because they have the same magnitude and direction, despite having different points as tip and tail.

As vectors in 2D space, both vectors $\overrightarrow{AB}$ and $\vec{u}$ can be described using components (x, y) .

A vector in a 3-dimensional space is depicted in the diagram below.



Vector $\vec{u}$ is a vector that spans across a space that is that can be represented by the x , y , and z axes.

Vectors in 3D space can be described using components (x, y, z) .

Component Form of a Vector

The component form of a vector refers to representing vectors in the form of (x, y) or (x, y, z) .

For a vector in n -space, then the component form would be $(a, b, c, \dots, n)$.

If a vector has a direction from point A to point B , and the coordinates of the points are $A(a_x, a_y, \dots, a_n)$ and $B(b_x, b_y, \dots, b_n)$, then the component form is determined by

$$x = b_x - a_x$$

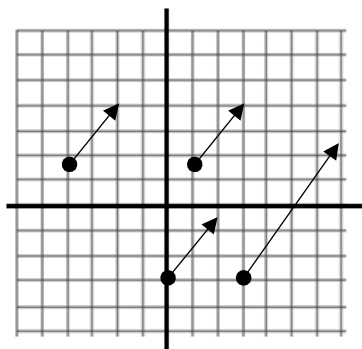
$$y = b_y - a_y$$

$$z = b_z - a_z$$

$$\vdots$$

$$n = b_n - a_n$$

Consider the following vectors $\vec{u}$, $\vec{v}$, $\vec{w}$, and $\vec{x}$, labelled from left to right, the following are the component forms.



It is noticeable that geometrically, vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ are the same. Vectors that are equal have the same magnitude and direction. They would also be the in same component form.

Vector $\vec{x}$ is a vector that is parallel to $\vec{u}$, $\vec{v}$, and $\vec{w}$. In component form, $\vec{x} = \langle 4, 4 \rangle = 2\langle 2, 2 \rangle = 2\vec{u}$.

Two vectors, $\vec{x}$ and $\vec{u}$ are parallel if $\vec{x} = k\vec{u}$ for k is a scalar.

Example 1

Determine the vector in component form if its tip is at the point $(1, 4, 0)$ and the tail is at the point $(2, 8, 9)$

$$\vec{u} = \langle 1 - 2, 4 - 8, 0 - 9 \rangle = \langle -1, -4, -9 \rangle$$

Example 2

Given the points $C(9, 5, 1)$ and $D(2, 3, 11)$, determine $\overrightarrow{CD}$ and $\overrightarrow{DC}$.

What can be concluded about these two vectors?

$$\overrightarrow{CD} = \langle 2 - 9, 3 - 5, 11 - 1 \rangle = \langle -7, -2, 10 \rangle$$

$$\overrightarrow{DC} = \langle 9 - 2, 5 - 3, 1 - 11 \rangle = \langle 7, 2, -10 \rangle$$

Notice that $\overrightarrow{CD} = -\overrightarrow{DC}$, the conclusion is that the two vectors are in opposition direction to one another.

Example 3

Determine the terminal point of a vector that is equivalent to $\vec{u} = \langle 1, 2 \rangle$ and has $A(1, 1)$ as its initial point.

Let B be the terminal point with coordinates (x, y) .

$$\vec{u} = (1, 2) = (x - 1, y - 1)$$

Solving for x and y would result in $x = 2$ and $y = 3$

So the terminal point of vector $\vec{u}$ is $(2, 3)$

Algebraic Properties of Vector Quantities in Component Form

If $\vec{u}$, $\vec{v}$, $\vec{w}$ are vectors in R^n (n -space) and if k and m are scalars, then the following are algebraic properties for operations of vectors:

- | | |
|--|---|
| a) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ | Commutative property |
| b) $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ | Associative property |
| c) $\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$ | Where $\vec{0}$ represents the zero vector $(0, 0, \dots, 0)$ |
| d) $\vec{u} + (-\vec{u}) = \vec{0}$ | Where $\vec{0}$ represents the zero vector $(0, 0, \dots, 0)$ |
| e) $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$ | Distributive property with scalar |
| f) $(k + m)\vec{u} = k\vec{u} + m\vec{u}$ | Distributive property with scalar |
| g) $k(m\vec{u}) = (km)\vec{u}$ | Associative property with scalar |
| h) $1\vec{u} = \vec{u}$ | Where 1 is a scalar |
| i) $0\vec{u} = \vec{0}$ | Where 0 is a scalar |
| j) $(-1)\vec{u} = -\vec{u}$ | Where -1 is a scalar |

Example 4

Given $\vec{u} = (1, 7, -6)$ and $\vec{v} = (0, -3, 8)$, determine the component vector of the following

- a) $8(\vec{u} - 2\vec{v})$
 b) $-\vec{u} + (2\vec{v} + \vec{u})$

a) $8(\vec{u} - 2\vec{v})$
 $\rightarrow 8((1, 7, -6) - 2(0, -3, 8))$
 $\rightarrow 8(1, 13, -22)$
 $\rightarrow (8, 104, -176)$

b) $-\vec{u} + (2\vec{v} + \vec{u})$
 $\rightarrow -(1, 7, -6) + (2(0, -3, 8) + (1, 7, -6))$

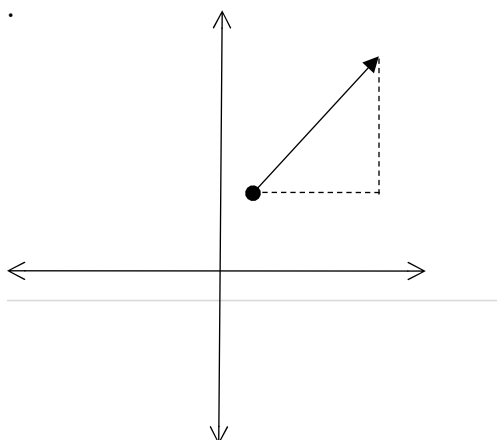
$$\begin{aligned} & \hat{i}(-1, -7, 6) + (1, 1, 10) \\ & \hat{i}(0, -6, 16) \end{aligned}$$

Practice

- 1) Given points $M(-9, 4, 12)$ and $N(0, 4, -6)$, determine $\overrightarrow{MN} \wedge \overrightarrow{NM}$.
- 2) Determine whether the given vectors are parallel to one another: $\vec{u} = (3, 9, 24)$ and $\vec{v} = (1, 3, 8)$. If so, determine the value of k for $\vec{u} = k \vec{v}$
- 3) Determine whether the given vectors are parallel to one another: $\vec{u} = (6, 18, -12)$ and $\vec{w} = (5, 15, -10)$. If so determine the value of k for $\vec{u} = k \vec{w}$
- 4) Determine whether the given vectors are in opposite directions to one another:
 $\vec{u} = (-76, 11, -2)$ and $\vec{w} = (76, -11, 2)$
- 5) Determine whether the given vectors in 4-D space are in opposite directions to one another:
 $\vec{u} = (6, -1, -5, 7)$ and $\vec{w} = (6, 1, 5, -7)$
- 6) Determine the initial point of $\overrightarrow{AB} = (1, 3, -4)$ if its terminal point is at $B(5, -6, -12)$
- 7) Given the following vectors, complete the following operation.
 $\vec{u} = (-9, 4, 10)$, $\vec{v} = (8, 12, -2)$ and $\vec{w} = (7, -1, 0)$
 - a) $5\vec{u} + 6\vec{v}$
 - b) $7\vec{u} + (\vec{w} - \vec{v})$
 - c) $-2(\vec{u} + \vec{v})$
 - d) $(\vec{v} - \vec{u}) + \vec{w}$
 - e) $(5+9)\vec{u} - 0\vec{v} + \vec{w}$

5.2 - The Norm of a Vector

The **norm** of a vector is the magnitude or length of the vector. The norm of a vector $\vec{u}$ will be $\|\vec{u}\|$, which is a scalar.



Consider the following vector,
 Suppose the initial and terminal points are P_1 and P_2 , respectively. Constructing a right-angled triangle can help determine the length of the vector.
 Then the sides of the triangle would be a and b and using Pythagorean Theorem,

In R^3 (3D space), the norm of a vector would be $|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

In R^n (n-space), the norm of a vector would be

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 + \dots + (n_2 - n_1)^2}$$

Every vector has a **unit vector**. The unit vector of a vector $\vec{u}$ is one that has the same direction as $\vec{u}$ but has a magnitude of 1 unit. The unit vector of $\vec{u}$ can be determined by dividing by its own magnitude.

The unit vector of $\vec{u}$:

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|}$$

Finding the unit vector of a vector is also known as **normalizing the vector**.

The following are examples of unit vectors:

$$(1, 0)$$

$$(1, 0, 0)$$

$$(0, 1, 0)$$

$$(0, 0, 1)$$

$$(1, 0, 0, 0)$$

$$(0, 1, 0, 0, 0)$$

The unit vectors, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ are specifically unit vectors that form the basis of the 3D space, R^3 . These are vectors that are along the x , y , z axes and have a magnitude of 1. They are known as **standard unit vectors** and commonly represented as:

$$\hat{i} = (1, 0, 0)$$

$$\hat{j} = (0, 1, 0)$$

$$\hat{k} = (0, 0, 1)$$

Every vector in R^3 can be written as a **linear combination** with the above unit vectors. This means that a vector $\vec{u}$ in 3D space can be written as the sum of the scalar products with the unit vectors of R^3

$$\vec{u} = a\hat{i} + b\hat{j} + c\hat{k}$$

where a, b, c are scalars

In fact, this is true in any n-space. Every vector in R^n can be written as a linear combination with unit vectors that form the basis on the n-space.

Example 1

Determine the norm of the following vector. $\vec{u} = (4, -2, 1)$

$$|\vec{u}| = \sqrt{4^2 + (-2)^2 + 1^2}$$

$$|\vec{u}| = \sqrt{16 + 4 + 1}$$

$$|\vec{u}| = \sqrt{21}$$

$$|\vec{u}| \approx 4.58$$

Example 2

Determine the unit vector of $\vec{v} = (3, 1, -9)$

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{3, 1, -9}{\sqrt{21}}$$

$$\hat{v} = \frac{1}{\sqrt{21}}, \frac{-9}{\sqrt{21}}$$

Example 3

Normalize the vector $\vec{v} = (3, 1, -9)$

Normalizing a vector means to determine its unit vector, which was completed in Example 2.

$$\hat{v} = \frac{1}{\sqrt{21}}, \frac{-9}{\sqrt{21}}$$

Example 4

Express the following vector $\vec{w} = (4, -6, 5, 10)$ in R^4 as a linear combination of its standard unit vectors

$$\vec{w} = 4(1, 0, 0, 0) - 6(0, 1, 0, 0) + 5(0, 0, 1, 0) + 10(0, 0, 0, 1)$$

Example 5

Determine the distance between the two points $M(1, 3, 4)$ and $N(-2, 4, 8)$

First, find the vector $\overrightarrow{MN}$

$$\overrightarrow{MN} = (-2 - 1, 4 - 3, 8 - 4) = (-3, 1, 4)$$

The distance between the two points would be equivalent to the length of the vector.

Next, find the norm of $\overrightarrow{MN}$

$$|\overrightarrow{MN}| = \sqrt{(-3)^2 + 1^2 + 4^2}$$

$$|\overrightarrow{MN}| = \sqrt{9 + 1 + 16}$$

$$|\overrightarrow{MN}| = \sqrt{26}$$

The distance between points M and N is $\sqrt{26}$.

Practice

1) Normalize the following vectors

a) $\vec{a} = (6, 5, 8)$

b) $\vec{b} = (3, 0, -11, 5)$

c) $\vec{c} = (0, 1, 0, 0, 0, 0)$

d) $\vec{d} = (-3, -17, -4)$

2) Is the following vector a unit vector? $\vec{p} = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$

3) Express the following vectors as a linear combination with its standard unit vectors

a) $\vec{v} = (5, -7, -9)$

b) $\vec{u} = (1, 0, 2, 5, -12)$

c) $\vec{m} = (-4, 1, 9, -3)$

4) Determine the distance between the given pairs of points

a) $M(1,2)$ and $N(9,-6)$

b) $A(1,5,4)$ and $B(0,2,-2)$

c) $P(1,4,5,-9)$ and $Q(4,5,6,-10)$

Unit 6: Vector Geometry

6.1 - The Dot Product

The concept of perpendicular vectors depends largely on a concept known as the dot product, which can be used to determine whether two vectors are perpendicular.

The **dot product** is defined as follows:

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$

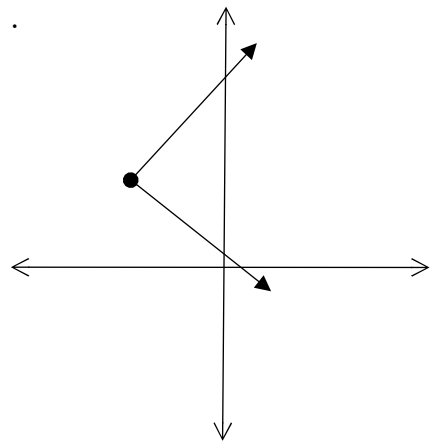
where θ is the angle between the vectors $\vec{u}$ and $\vec{v}$ when they meet at their initial points.

If $\vec{u} = (u_1, u_2, \dots, u_n)$ and $\vec{v} = (v_1, v_2, \dots, v_n)$, then using component forms of the vectors, the dot product can also be found as follows:

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

The dot product of two vectors is a scalar.

When two vectors are perpendicular to one another, the angle between them would be 90° .



From the diagram, vectors $\vec{u}$ and $\vec{v}$ are perpendicular to one another.

Their dot product would be:

Since

Thus, using the dot product can help determine whether two vectors are perpendicular to one another. If the dot product is 0, then the conclusion is that they are perpendicular.

Example 1

Determine the dot product of the following vectors, given their component form.

$$\vec{u} = (2, -1, -2)$$

$$\vec{v} = (3, 1, -9)$$

Using the formula: $\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

$$\vec{u} \cdot \vec{v} = (2)(3) + (-1)(1) + (-2)(-9)$$

$$\vec{u} \cdot \vec{v} = 6 - 1 + 18$$

$$\vec{u} \cdot \vec{v} = 23$$

Example 2

Determine the dot product of vectors $\vec{a}$ and $\vec{b}$ given the following information: $\vec{a} = (1, 2, 0)$, $|\vec{b}| = 5$, and the angle between them is 25°

State whether the vectors are perpendicular to one another.

In this case, we cannot use the formula: $\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

$$\text{So } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

First, we need to determine $|\vec{a}|$

$$|\vec{a}| = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{5}$$

$$\vec{a} \cdot \vec{b} = (\sqrt{5})(5) \cos 25^\circ$$

$$\vec{a} \cdot \vec{b} \approx 10.13$$

Since their dot product is not zero, therefore, vectors $\vec{a}$ and $\vec{b}$ are not perpendicular to one another.

Example 3

Given $\vec{u} = (4, 2, 7)$ and $\vec{v} = (-4, -2, 0)$, determine the angle between the vectors.

Since $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$, we can use the dot product to determine θ

$$\vec{u} \cdot \vec{v} = (4)(-4) + (2)(-2) + (7)(0) = -20$$

$$|\vec{u}| = \sqrt{4^2 + 2^2 + 7^2} = \sqrt{69}$$

$$|\vec{v}| = \sqrt{(-4)^2 + (-2)^2 + 0^2} = \sqrt{20}$$

Rearranging to solve for θ

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$$

$$\theta = \cos^{-1} \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \right)$$

$$\theta = \cos^{-1} \left(\frac{-20}{\sqrt{69} \sqrt{20}} \right)$$

$$\theta = 106.85^\circ \text{ or } 253.15^\circ$$

Algebraic Properties of the Dot Product

If $\vec{u}$, $\vec{v}$, $\vec{w}$ are vectors in R^2 or R^3 and k is a scalar, then the following are algebraic properties of for the dot product:

- | | |
|--|--|
| a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$ | Commutative property |
| b) $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ | Distributive property |
| c) $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$ | Distributive property |
| d) $k(\vec{u} \cdot \vec{v}) = (k\vec{u}) \cdot \vec{v} = \vec{u} \cdot (k\vec{v})$ | Associative property |
| e) $\vec{u} \cdot \vec{0} = \vec{0} \cdot \vec{u} = 0$ | Where $\vec{0}$ represents the zero vector |
| f) $\vec{u} \cdot \vec{u} = \vec{u} ^2$ | |

Example 4

Using algebraic properties of the dot product, determine $|\vec{u} + \vec{v}|^2$ given $|\vec{u}| = 3$, $|\vec{v}| = 5$ and $\vec{u} \cdot \vec{v} = 9$

According to property f, the dot product of a vector and itself results in the square of its magnitude.

$$|\vec{u} + \vec{v}|^2 = (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v})$$

$$|\vec{u} + \vec{v}|^2 = \vec{u} \cdot \vec{u} + \vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{u} + \vec{v} \cdot \vec{v}$$

According to the commutative property, $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

$$|\vec{u} + \vec{v}|^2 = |\vec{u}|^2 + 2(\vec{u} \cdot \vec{v}) + |\vec{v}|^2$$

$$|\vec{u} + \vec{v}|^2 = 3^2 + 2(9) + 5^2$$

$$|\vec{u} + \vec{v}|^2 = 9 + 18 + 25$$

$$|\vec{u} + \vec{v}|^2 = 52$$

Practice

- 1) Determine the dot product of the following vectors
 - a) $\vec{p} = (4, 3, 1)$ and $\vec{q} = (4, 1, -10)$
 - b) $\vec{v} = (9, 1, 18)$ and $\vec{w} = (-3, -1, 0)$
 - c) $|\vec{a}| = 6$, $|\vec{c}| = 7$, and the angle between them is 82°

- 2) Determine whether the following vectors are perpendicular
 - a) $\vec{m} = (1, 3, 1)$ and $\vec{q} = (4, 1, -7)$
 - b) $\vec{p} = (0, -3, 4)$ and $\vec{q} = (6, 8, 6)$
 - c) $\vec{c} = (14, 13, -1, 5)$ and $\vec{d} = (-4, 2, -3, 5)$

- 3) Given $\vec{u}$ and $\vec{v}$ are parallel to one another, show that the dot product $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}|$

- 4) Given $\vec{u} = (1, y, -6)$ and $\vec{v} = (4, -2, -1)$, determine the value of y so that $\vec{u}$ and $\vec{v}$ are perpendicular.

- 5) Given $\vec{u} = (9, 1, -2)$ and $\vec{v} = (4, 0, 5)$, determine the angle between the vectors.

- 6) Using algebraic properties, determine $|2\vec{u} - 3\vec{v}|^2$ given $|\vec{u}| = 6$, $|\vec{v}| = 5$ and $\vec{u} \cdot \vec{v} = -12$

- 7) Given $\vec{p} = (4, 3, 1)$ and $\vec{q} = (4, 1, -10)$, verify the commutative property of the dot product.

6.2 - Orthogonality

In Chapter 6.1, the dot product and its relation to perpendicular vectors were discussed.

Recall that if $\vec{u} \cdot \vec{v} = 0$, then the angle between the vectors is 90° , which means that $\vec{u}$ and $\vec{v}$ are perpendicular vectors. In vector geometry, they are also known as orthogonal vectors.

This chapter will focus on exploring the concept of orthogonality further and applications of the dot product on orthogonality.

Orthogonal Sets

Consider the following vectors:

$$\vec{u} = (-2, 1, 1) \quad \vec{v} = (1, 0, 2) \quad \vec{w} = (-2, -5, 1)$$

These vectors are **mutually orthogonal**, which means that they are orthogonal to one another.

The dot product will be calculated to verify that they are mutually orthogonal:

$$\vec{u} \cdot \vec{v} = (-2)(1) + (1)(0) + (1)(2) = 0$$

$$\vec{u} \cdot \vec{w} = (-2)(-2) + (1)(-5) + (1)(1) = 0$$

$$\vec{v} \cdot \vec{w} = (1)(-2) + (0)(-5) + (2)(1) = 0$$

When all pairs of distinct vectors in a set of vectors are orthogonal, the set of vectors is known as an orthogonal set.

For example, the x, y, z axes make up an orthogonal set of vectors as they are mutually perpendicular to one another.

Orthonormal Sets

An orthonormal set of vectors is an orthogonal set that contains only unit vectors.

This means that not only are pairs of distinct vectors in the set orthogonal, they also each has a magnitude of 1.

For example, the standard unit vectors in $\mathbb{R}^3$, $\hat{i}$, $\hat{j}$, and $\hat{k}$ make up an orthonormal set.

Example 1

Determine whether the following vectors belong in an orthogonal set in $\mathbb{R}^3$

$$\vec{a} = (1, 3, -9)$$

$$\vec{b} = (0, 3, 1)$$

$$\vec{c} = (1, 1, -3)$$

To determine whether the vectors are mutually orthogonal, the dot product needs to be used.

$$\vec{a} \cdot \vec{b} = (1)(0) + (3)(3) + (-9)(1) = 0$$

$$\vec{b} \cdot \vec{c} = (0)(1) + (3)(1) + (1)(-3) = 0$$

$$\vec{a} \cdot \vec{c} = (1)(1) + (3)(1) + (-9)(-3) = 31$$

The dot products show that vectors $\vec{a}$ is orthogonal to $\vec{b}$, and $\vec{b}$ is orthogonal to $\vec{c}$. However, the pair of vectors, $\vec{a}$ and $\vec{c}$ are not orthogonal to one another. Therefore, this is not an orthogonal set.

Example 2

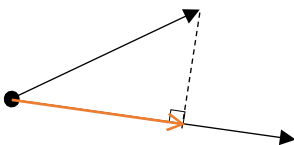
Give an example of an orthonormal set in R^5

The most convenient orthonormal set would be the set of standard unit vectors of R^5 . These are:

$$(1, 0, 0, 0, 0), (0, 1, 0, 0, 0), (0, 0, 1, 0, 0), (0, 0, 0, 1, 0), \text{ and } (0, 0, 0, 0, 1)$$

Projections

A **vector projection** of vector $\vec{u}$ onto a non-zero vector $\vec{v}$ is the orthogonal projection of $\vec{u}$ onto a line parallel to vector $\vec{v}$. In the diagram below, the orange vector is the result of projecting $\vec{u}$ onto $\vec{v}$. This vector is known as the $\text{proj}_{\vec{v}} \vec{u}$, which is read as the “projection on $\vec{u}$ onto $\vec{v}$ ”.



To determine the projection vector, consider the angle between vectors $\vec{u}$ and $\vec{v}$ to be θ .

Since vector projection is an orthogonal projection, a right-angled triangle is formed.

The length of the vector $\text{proj}_{\vec{v}} \vec{u}$ can be calculated using trigonometry:

$$\cos \theta = \frac{|\text{proj}_{\vec{v}} \vec{u}|}{|\vec{u}|}$$

Recall that the formula of the dot product can be rearranged to:

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$$

So

$$\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} = \frac{|\text{proj}_{\vec{v}} \vec{u}|}{|\vec{u}|}$$

Rearranging results in the following formula to calculate the magnitude of the projection vector:

$$|\text{proj}_{\vec{v}} \vec{u}| = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$$

Since the projection vector is parallel to vector $\vec{v}$, then the $\text{proj}_{\vec{v}} \vec{u}$ would have the magnitude of the $\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$ and the direction of $\vec{v}$. Using the unit vector, $\hat{v}$, since it has a magnitude of 1 and the same direction as $\vec{v}$, the projection vector can be written as:

$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \hat{v}$$

$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \left(\frac{\vec{v}}{|\vec{v}|} \right)$$

Thus, the formula to calculate the projection of $\vec{u}$ onto $\vec{v}$ is:

$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}$$

Example 3

Determine the projection of $\vec{u}$ onto $\vec{v}$ given $\vec{u} = (7, 2, 1)$ and $\vec{v} = (3, 0, -1)$

$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}$$

$$\text{proj}_{\vec{v}} \vec{u} = \frac{(7)(3) + (2)(0) + (1)(-1)}{(\sqrt{3^2 + 0^2 + (-1)^2})^2} \vec{v}$$

$$\text{proj}_{\vec{v}} \vec{u} = \frac{20}{10} \vec{v}$$

$$\text{proj}_{\vec{v}} \vec{u} = 2 \vec{v}$$

$$\text{proj}_{\vec{v}} \vec{u} = 2(3, 0, -1)$$

$$\text{proj}_{\vec{v}} \vec{u} = (6, 0, -2)$$

Practice

- 1) Determine whether the following set of vectors is an orthogonal set.
 - a) $(1, -1, 0), (1, 1, 0), (0, 0, 1)$
 - b) $(6, 7, 2), (5, 0, -15), (0, 1, 12)$
 - c) $(1, 1, 0, 3), (4, -4, 7, 0), (-2, 2, 2, 0), (1, 9, -2, 1)$

- 2) Determine whether the following set is an orthonormal set

$$(1, 0, -1), (1, \sqrt{2}, 1), (1, -\sqrt{2}, 1)$$

- 3) Give an example of an orthonormal set in $\mathbb{R}^4$

- 4) Determine the value of x such that the following vectors are orthogonal

$$(x, -2, 1) \text{ and } (2, -1, 3)$$

- 5) Determine the projection of $\vec{u}$ onto $\vec{v}$ given the following vectors
 - a) $\vec{u} = (5, 7, 1)$ and $\vec{v} = (2, -1, 3)$
 - b) $\vec{u} = (3, -2, 1)$ and $\vec{v} = (4, 1, 1)$
 - c) $\vec{u} = (9, 0, -12)$ and $\vec{v} = (6, -19, -3)$

6.3 - The Cross Product

For vectors in R^3 , the **cross product** provides a systematic way to determine a vector that is mutually orthogonal to two other vectors.

Given vectors $\vec{u} = (x_1, y_1, z_1)$ and $\vec{v} = (x_2, y_2, z_2)$, then a vector orthogonal to $\vec{u}$ and $\vec{v}$ can be determined by using the cross product, which is defined as:

$$\vec{u} \times \vec{v} = (y_1 z_2 - z_1 y_2, z_1 x_2 - x_1 z_2, x_1 y_2 - y_1 x_2)$$

Another form of this formula is to use the determinant:

$$\vec{u} \times \vec{v} = \det \begin{bmatrix} y_1 & y_2 \\ z_1 & z_2 \end{bmatrix} \hat{i} - \det \begin{bmatrix} x_1 & x_2 \\ z_1 & z_2 \end{bmatrix} \hat{j} + \det \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \hat{k}$$

There is a method to remembering the determinant form of the formula as well:

First, write the two vectors' components across and in two rows

$$x_1 y_1 z_1 \quad x_1 y_1 z_1$$

$$x_2 y_2 z_2 \quad x_2 y_2 z_2$$

Next, disregard the column 1 and 6 and focus on column 2-5. The three components of $\vec{u} \times \vec{v}$ are:

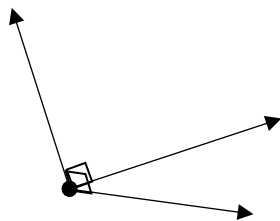
$$\begin{vmatrix} x_1 y_1 z_1 & x_1 y_1 z_1 \\ x_2 y_2 z_2 & x_2 y_2 z_2 \end{vmatrix}$$

x is the result from taking the determinant of the matrix formed by column 2 and 3

y is the result from taking the determinant of the matrix formed by column 3 and 4

z is the result from taking the determinant of the matrix formed by column 4 and 5

Unlike the dot product, the result of the cross product is a vector. The following diagram shows the vector $\vec{u} \times \vec{v}$, which is a vector orthogonal to both $\vec{u}$ and $\vec{v}$. Note that the three vectors are not necessarily an orthogonal set because $\vec{u}$ and $\vec{v}$ are not necessarily orthogonal to one another.



Example 1

Determine the cross product of the following vectors: $\vec{u} = (1, 2, -6)$ and $\vec{v} = (3, 1, -9)$

Using the determinant method

$$x = \det \begin{bmatrix} 2 & -6 \\ 1 & -9 \end{bmatrix} = (2)(-9) - (1)(-6) = -12$$

$$y = \det \begin{bmatrix} -6 & 1 \\ -9 & 3 \end{bmatrix} = (-6)(3) - (-9)(1) = -9$$

$$z = \det \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} = (1)(1) - (3)(2) = -5$$

The cross product, $\vec{u} \times \vec{v} = (-12, -9, -5)$

Example 2

Determine the vector that is orthogonal to both of the following vectors:

$\vec{a} = (7, -2, 1)$ and $\vec{b} = (-5, 0, 4)$

Verify the orthogonality.

We need to determine $\vec{a} \times \vec{b}$

$$\begin{array}{ccc} 7 & -2 & 1 \\ -5 & 0 & 4 \end{array}$$

$$x = \det \begin{bmatrix} -2 & 1 \\ 0 & 4 \end{bmatrix} = -8$$

$$y = \det \begin{bmatrix} 1 & 7 \\ 4 & -5 \end{bmatrix} = \textcolor{red}{-8} - 33 \textcolor{red}{-10}$$

$$z = \det \begin{bmatrix} 7 & -2 \\ -5 & 0 \end{bmatrix} = \textcolor{red}{-10} - 10 \textcolor{red}{-10}$$

Let vector $\vec{c} = \vec{a} \times \vec{b}$

$$\vec{c} = (-8, -33, -10)$$

To verify the orthogonality, the dot product will be used:

$$\vec{a} \cdot \vec{c} = (7)(-8) + (-2)(-33) + (1)(-10) = 0$$

$$\vec{b} \cdot \vec{c} = (-5)(-8) + (0)(-33) + (4)(-10) = 0$$

This shows that vector $\vec{c}$ is orthogonal to both $\vec{a}$ and $\vec{b}$.

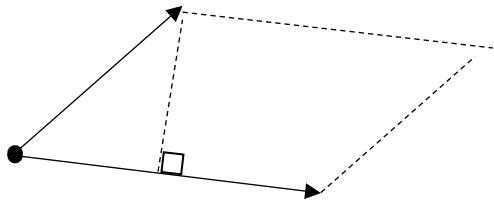
The Magnitude of $\vec{u} \times \vec{v}$

The magnitude of the vector obtained from the cross product can be found using the components of the vector: $\sqrt{x^2 + y^2 + z^2}$

Another method is to use the following formula:

$$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta$$

The application of calculating $|\vec{u} \times \vec{v}|$ is in finding the area of a parallelogram formed by $\vec{u}$ and $\vec{v}$.



The area of a parallelogram is the product of base and height.

$$A = h |\vec{v}|$$

Using trigonometry, $h = |\vec{u}| \sin \theta$

$$A = (|\vec{u}| \sin \theta) |\vec{v}|$$

$$A = |\vec{u}| |\vec{v}| \sin \theta$$

$$A = |\vec{u} \times \vec{v}|$$

Example 3

Determine the area of the parallelogram formed by $\vec{a} = (7, -2, 1)$ and $\vec{b} = (-5, 0, 4)$

Using the cross product, $\vec{a} \times \vec{b}$ is determined to be $(-8, -33, -10)$

$$|\vec{a} \times \vec{b}| = \sqrt{(-8)^2 + (-33)^2 + (-10)^2}$$

$$|\vec{a} \times \vec{b}| = \sqrt{1253}$$

$$|\vec{a} \times \vec{b}| \approx 35.40$$

Example 4

Using the cross product, determine the angle between $\vec{u} = (5, 6, 1)$ and $\vec{v} = (6, -1, 2)$

Using the cross product, $\vec{u} \times \vec{v} = (13, -4, -41)$

$$|\vec{u} \times \vec{v}| = \sqrt{13^2 + (-4)^2 + (-41)^2} = \sqrt{1866}$$

$$|\vec{u}| = \sqrt{5^2 + 6^2 + 1^2} = \sqrt{62}$$

$$|\vec{v}| = \sqrt{6^2 + (-1)^2 + 2^2} = \sqrt{41}$$

Since $|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta$,

$$\sin \theta = \frac{|\vec{u} \times \vec{v}|}{|\vec{u}| |\vec{v}|}$$

$$\sin \theta = \frac{\sqrt{1866}}{\sqrt{62} \sqrt{41}}$$

$$\theta \approx 58.96^\circ$$

Algebraic Properties of the Cross Product

If $\vec{u}$, $\vec{v}$, $\vec{w}$ are vectors in R^3 and k is a scalar, then the following are algebraic properties of for the cross product:

- | | |
|---|--|
| a) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$ | The cross product is not commutative |
| b) $\vec{u} \times (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) + (\vec{u} \times \vec{w})$ | Distributive property |
| c) $(\vec{u} + \vec{v}) \times \vec{w} = (\vec{u} \times \vec{w}) + (\vec{v} \times \vec{w})$ | Distributive property |
| d) $k(\vec{u} \times \vec{v}) = (k\vec{u}) \times \vec{v} = \vec{u} \times (k\vec{v})$ | Associative property |
| e) $\vec{u} \times \vec{0} = \vec{0} \times \vec{u} = \vec{0}$ | Where $\vec{0}$ represents the zero vector |
| f) $\vec{u} \times \vec{u} = \vec{0}$ | Where $\vec{0}$ represents the zero vector |

Practice

- 1) Determine the cross product of the following vectors
 - a) $\vec{u} = (1, 8, 5)$ and $\vec{v} = (-3, -12, 5)$
 - b) $\vec{p} = (3, -2, -6)$ and $\vec{q} = (0, 7, -4)$

- 2) Determine the vector that is orthogonal to both of the following vectors. Verify the orthogonality
 - a) $\vec{u} = (4, 12, 9)$ and $\vec{v} = (1, -1, -10)$
 - b) $\vec{a} = (-1, 14, 2)$ and $\vec{b} = (7, -1, 0)$

- 3) Determine the area of the parallelogram formed by the following vectors
 - a) $\vec{u} = (4, -2, 1)$ and $\vec{v} = (2, 5, -6)$
 - b) $\vec{m} = (0, 3, 7)$ and $\vec{n} = (5, -2, 11)$

- 4) Determine the area of the parallelogram with the following points as three of its vertices
 $A(6, 3, 5), B(1, 3, 0), C(1, -7, 2)$

- 5) Using the cross product, determine the angle between $\vec{m} = (1, -3, 4)$ and $\vec{n} = (-5, 5, 8)$

- 6) Using the given vectors, $\vec{u} = (4, -2, 1)$ and $\vec{v} = (2, 5, -6)$, verify the following two algebraic properties
 - a) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$
 - b) $\vec{u} \times \vec{u} = \vec{0}$

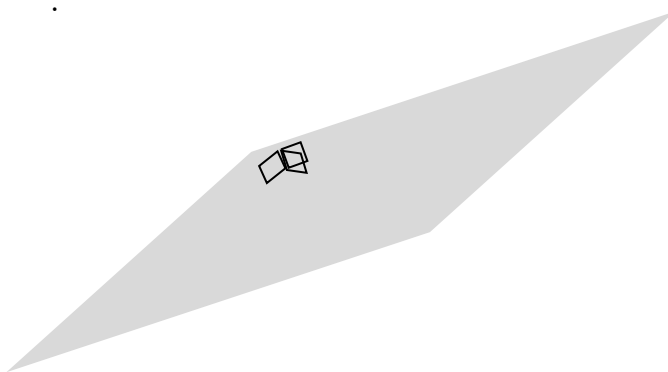
6.4 - The Triple Scalar Product and Coplanar Vectors

The **triple scalar product** is defined as the following:

$$\vec{w} \cdot (\vec{u} \times \vec{v})$$

The operation of the cross product is completed first, resulting in a vector. Then the dot product of vector $\vec{u}$ and the vector resulted from the cross product is completed. The result of the dot product is a scalar. Therefore, the result of the triple scalar product is a scalar, as the name suggests.

The application of the triple scalar product is behind the following concept:



Recall that $\vec{u} \times \vec{v}$ is the vector that is orthogonal to both $\vec{u}$ and $\vec{v}$.

If $\vec{w} \cdot (\vec{u} \times \vec{v})$ results in 0, then $\vec{w}$ is orthogonal to $\vec{u} \times \vec{v}$.

This also means that $\vec{w}$, $\vec{u}$, and $\vec{v}$ lie on the same plane.

It is said that $\vec{w}$, $\vec{u}$, and $\vec{v}$ are **coplanar**.

Example 1

Determine whether the following vectors are coplanar: $\vec{u} = (1, 0, -2)$, $\vec{v} = (2, 1, 1)$, $\vec{w} = (4, 0, 2)$

We need to determine the triple scalar product

$$\vec{u} \times \vec{v} = (0+2, -4-1, 1-0) = (2, -5, 1)$$

$$\vec{w} \cdot (\vec{u} \times \vec{v}) = (4, 0, 2) \cdot (2, -5, 1)$$

$$\vec{w} \cdot (\vec{u} \times \vec{v}) = (4)(2) + (0)(-5) + (2)(1)$$

$$\vec{w} \cdot (\vec{u} \times \vec{v}) = 10$$

Since $\vec{w} \cdot (\vec{u} \times \vec{v}) \neq 0$, then the vectors are not coplanar.

Example 2

Determine the value of z such that the three vectors would be coplanar:

$$\vec{a} = (9, 0, z), \vec{b} = (2, -1, 5), \vec{c} = (8, 3, -2)$$

For the vectors to be coplanar, then $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$

$$\vec{b} \times \vec{c} = (2-15, 40+4, 6+8)$$

$$\vec{b} \times \vec{c} = (-13, 44, 14)$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (9)(-13) + (0)(44) + (z)(14) = 0$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = -117 + 14z = 0$$

$$z = \frac{117}{14}$$

Algebraic Properties of the Triple Scalar Product and Operations with Both the Dot and Cross Products

If $\vec{u}$, $\vec{v}$, $\vec{w}$ are vectors in R^3 , then the following are algebraic properties of for the triple scalar product and operations that involve both the dot and cross product:

$$a) \quad \vec{w} \cdot (\vec{u} \times \vec{v}) = \vec{u} \cdot (\vec{w} \times \vec{v}) = \vec{v} \cdot (\vec{w} \times \vec{u})$$

$$b) \quad \vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{u} \cdot \vec{v})\vec{w}$$

$$c) \quad (\vec{u} \times \vec{v}) \times \vec{w} = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{v} \cdot \vec{w})\vec{u}$$

Example 3

Given the following vectors, verify the property $(\vec{u} \times \vec{v}) \times \vec{w} = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{v} \cdot \vec{w})\vec{u}$

$$\vec{u} = (2, 1, -2), \vec{v} = (6, 2, 1), \vec{w} = (0, 0, -7)$$

$$\vec{u} \times \vec{v} = (5, -14, -2)$$

$$(\vec{u} \times \vec{v}) \times \vec{w} = (98, 35, 0)$$

$$\vec{u} \cdot \vec{w} = 14$$

$$\vec{v} \cdot \vec{w} = -7$$

$$(\vec{u} \cdot \vec{w})\vec{v} - (\vec{v} \cdot \vec{w})\vec{u} = 14(6, 2, 1) - (-7)(2, 1, -2) = (98, 35, 0)$$

$$(\vec{u} \times \vec{v}) \times \vec{w} = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{v} \cdot \vec{w})\vec{u} = (98, 35, 0)$$

Practice

- 1) Determine the triple scalar product for the following vectors

$$\vec{u} = (8, 2, -2), \vec{v} = (1, 5, 0), \vec{w} = (4, 1, -3)$$

- 2) Determine whether the following vectors are coplanar

a) $\vec{u} = (1, 5, -2), \vec{v} = (3, -1, 0), \vec{w} = (5, 9, -4)$

b) $\vec{a} = (3, 1, -2), \vec{b} = (1, 0, -2), \vec{c} = (5, 1, 10)$

c) $\vec{m} = (7, 9, -1), \vec{n} = (1, 2, -8), \vec{p} = (6, 0, -2)$

- 3) Determine the value of y such that the three vectors would be coplanar

$$\vec{c} = (5, 7, 3), \vec{d} = (-1, y, 2), \vec{f} = (5, 1, 1)$$

- 4) Determine the value of x such that the three vectors would be coplanar

$$\vec{a} = (x, 4, 2), \vec{b} = (1, 2, -9), \vec{c} = (1, -1, 6)$$

- 5) Determine the value of x such that the three vectors would be coplanar

$$\vec{u} = (x, 1, 3), \vec{v} = (x, 4, 1), \vec{w} = (-2, 0, 2)$$

- 6) Given the vectors, verify the algebraic properties

$$\vec{u} = (0, -2, 5), \vec{v} = (4, 1, 6), \vec{w} = (-6, 2, 9)$$

$$a) \quad \vec{w} \cdot (\vec{u} \times \vec{v}) = \vec{u} \cdot (\vec{w} \times \vec{v}) = \vec{v} \cdot (\vec{w} \times \vec{u})$$

$$b) \quad \vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w}) \vec{v} - (\vec{u} \cdot \vec{v}) \vec{w}$$

Lab Activities

Introduction to SciLab

SciLab is an open source mathematical software that is used for computations. The software can be downloaded from www.scilab.org.

In the Centennial College computer labs, the software can be accessed under “More Programs”.

In this Introduction section, we will work on the basic commands for inputs and operations.

File Management:

Action	Command	Input Example
Saving a file	Save ('name.dat')	Save('lab1.dat')
Loading a file	Load('name.dat')	Load('lab2.dat')

Inputting Basic Operations:

Operation	Command	Input Example
Addition	+	2+2
Subtraction	-	4-3
Multiplication	*	7*5
Division	/	60/10

Exponents	$\wedge$	2^3
Square Roots	Sqrt()	Sqrt(16)
Rounding	Round()	Round(1.6)
Rounding up to the nearest integer	Ceil()	Ceil(2.1)
Rounding down to the nearest integer	Floor()	Floor(3.4)
Defining variables	=	A=5

Inputting a matrix:

- Use square brackets “[]” to mark the beginning and the end of a matrix
- Use commas “,” to separate the values in different columns
- Use semicolons “;” to separate the values of different rows

To input a matrix A with size $m \times n$

$$A = [a_{11}, a_{12}, a_{13}, \dots, a_{1n}; a_{21}, a_{22}, a_{23}, \dots, a_{2n}; \dots; a_{m1}, a_{m2}, a_{m3}, \dots, a_{mn}]$$

E.g. To enter $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Input:

$$A = [1, 2; 3, 4]$$

Unit 1: Systems of Linear Equations

Example 1

Solve the following system of linear equations

$$\begin{aligned} a + 2b + c &= -3 \\ 2a - b + 3c &= 3 \\ -a + 5b + 5c &= -11 \end{aligned}$$

Using SciLab:

Action	Command
Inputting the matrix	$A = [1, 2, 1, -3; 2, -1, 3, 3; -1, 5, 5, -11]$
$\begin{bmatrix} 1 & 2 & 1 & -3 \\ 2 & -1 & 3 & 3 \\ -1 & 5 & 5 & -11 \end{bmatrix}$	
Compute the matrix to	rref(A)

be row-echelon form

SciLab Practice

Solve the following systems of linear equation. State the type of solution of the system.

1) $74 + 5x = -3y$
 $-3y = 26 + 2x$

2) $3x + 4y + z = 1$
 $2x + 3y = 0$
 $4x + 3y - z = -2$

3) $2x + 1 = -y - z$
 $x + 2y = -z$
 $3x = 5 + 2z$

Unit 2: Matrix Operations

The following are commands for matrices and matrix operations:

Operation	Command	Input Example
Addition	$+$	A+B
Subtraction	$-$	A-B
Multiplication	$*$	A*B
Exponents	$^$	2^3
Transpose	A'	A'
Display an entry a_{nm} in matrix A	A(n,m)	A(2,3)
Change an entry a_{nm} in matrix A	A(n,m)=k	A(2,3)=5
Display the entire n^{th} column	A(:,n)	A(:,2)
Display the entire n^{th} row	A(n,:)	A(3,:)
Compute size of matrix	size(A)	size(A)
Computes trace of a matrix	trace(A)	trace(A)
Creates the zero matrix size $n \times m$	zeros(n,m)	zeros(2,3)
Creates a matrix size $n \times m$ with 1's	ones(n,m)	ones(5,2)
Creates a $n \times n$ identity matrix	eye(n,n)	eye(2,2)

SciLab Practice

Given the following matrices, input them into SciLab.

Perform the following operations using SciLab and the commands introduced above.

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 7 & 0 & 2 \\ 2 & -5 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 & 1 \\ 15 & 1 & 5 \\ -1 & 4 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & -2 \\ 4 & 1 & 3 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 4 \\ 7 & 8 \\ 1 & 1 \end{bmatrix}$$

- 1) Compute the size of each of the matrices
- 2) Create a 3×3 Identity matrix
- 3) $AI + B$
- 4) $3B - 2A$
- 5) $C + D$
- 6) $D^T + 2C$
- 7) CD
- 8) DC
- 9) Is matrix multiplication commutative?
- 10) Compute the trace of B
- 11) $tr(A) + tr(B)$
- 12) $tr(A + B)$
- 13) $B^T A$
- 14) Change entry $c_{21} = 5$
- 15) Display row 1 in matrix D

Unit 3: Inverse of a Matrix

The command on SciLab to calculate the inverse of a matrix is: `Inv(A)`

Example 1

Given the following matrices, determine whether they are inverses of one another

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 2 \\ 1 & -2 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 4 & -2 & -5 \\ 2 & -1 & -3 \\ 1 & 0 & -1 \end{bmatrix}$$

- 1) Input the matrices. E.g. A[1, -2, 1; -1, 1, 2; 1, -2, 0]
- 2) Determine the product to see whether the product is the identity matrix

Example 2

Given the following matrix, using SciLab, calculate $ad - bc$ to test whether the matrix is invertible

$$A = \begin{bmatrix} 5 & 6 \\ 1 & 3 \end{bmatrix}$$

To calculate $ad - bc$, one way is to directly input the values and the operations
 $(5*3)-(1*6)$

or you can get SciLab to use the appropriate entries to perform $ad - bc$
 $A(1,1)*A(2,2)-A(2,1)*A(1,2)$

Example 3

Determine the inverse of the following matrix

$$A = \begin{bmatrix} 5 & 6 \\ 1 & 3 \end{bmatrix}$$

- 1) Input the matrix first: A[5, 3; 1, 6]
- 2) Use the command Inv(A) to calculate the inverse: Inv(A)

Example 4

Solve the following linear system using the inverse of a matrix

$$\begin{aligned} -2x + y + 3z &= 1 \\ 3x - y + z &= 5 \\ -5x + 2z &= -1 \end{aligned}$$

- 1) Input matrix A
- 2) Input matrix C
- 3) Determine A^{-1} using the inverse command

- 4) Determine the product $A^{-1}C$

SciLab Practice

- 1) Determine whether the following matrices are inverses of one another

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 0 \\ -1 & 1 & -5 \end{bmatrix}$$

- 2) Test whether the following matrix is invertible

$$C = \begin{bmatrix} 7 & 14 \\ -2 & -4 \end{bmatrix}$$

- 3) Determine the inverse of the following matrix

$$B = \begin{bmatrix} 8 & 2 & 1.5 \\ -2 & 3 & -7 \\ 12 & -1 & 1 \end{bmatrix}$$

- 4) Solve the following linear system using inverse of a matrix

$$\begin{aligned} a + 10c &= 5 \\ 3a + b - 4c &= -1 \\ 4a + b + 6c &= 1 \end{aligned}$$

Unit 4: Special Matrices and Determinants

For this unit, SciLab will be used to demonstrate the properties of the special matrices and to calculate determinants.

Example 1

Given $A = \begin{bmatrix} 7 & 1 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 8 \end{bmatrix}$, demonstrate the property that the transpose of an upper triangular

matrix results in a lower triangular matrix.

- 1) Input matrix A
- 2) Use the transpose command: A'

Example 2

Compute the determinant of the following matrix, $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 9 \\ 7 & 1 & 6 \end{bmatrix}$

- 1) Input matrix A
- 2) Use $\det(A)$

SciLab Practice

Given the following special matrices, use SciLab to demonstrate the following properties.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 10 & 1 \\ 0 & 0 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 7 & 0 \\ 0 & 9 \end{bmatrix} \quad E = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \\ 6 & 11 & 6 \end{bmatrix}$$

- 1) Using matrix A , demonstrate the property that a diagonal matrix is invertible if and only if all of its entries on the main diagonal are non-zero values.
- 2) Using matrix B , demonstrate the property that a triangular matrix is invertible if and only if all of its entries on the main diagonal are non-zero values

- 3) Using matrix A , demonstrate the property $A = \begin{bmatrix} a & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k \end{bmatrix}$, $A^{-1} = \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & k^{-1} \end{bmatrix}$ for

diagonal matrices

- 4) Using matrices A and C , demonstrate that $A+C$ and $A-C$ are also symmetric.
- 5) Using matrix B , demonstrate that the inverse of an invertible upper triangular matrix is an upper triangular matrix
- 6) Using matrix E , demonstrate that the inverse of an invertible lower triangular matrix is a lower triangular matrix
- 7) Using matrix E , demonstrate that the transpose of a lower triangular matrix is an upper

triangular matrix.

Unit 5: Vectors in 2, 3, and n-Space

The following are commands for inputting vectors on SciLab:

Action	Command	Input Example
Inputting a vector	$p=[x, y, z]$	$p=[1, 2, 3]$
Adding two vectors	+	$p+q$
Subtracting two vectors	-	$p-q$
Scalar multiplication	kp	$2p$
Computing norm of a vector	$\text{norm}(p)$	$\text{norm}(p)$

SciLab Practice

- Given the vectors $\vec{a}=(1, -9, 6)$, $\vec{b}=(2, 5, -1)$, and $\vec{c}=(3, 6, -7)$, use SciLab to demonstrate the following properties
 - $\vec{a}+\vec{b}=\vec{b}+\vec{a}$
 - $(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$
 - $\vec{c}+(-\vec{c})=\vec{0}$
- Given the vectors $\vec{a}=(1, -9, 6)$, $\vec{b}=(2, 5, -1)$, and $\vec{c}=(3, 6, -7)$, use SciLab to complete the following operation
 - $2(\vec{a}+\vec{b})$
 - $6\vec{a}+4\vec{b}$
 - $\vec{c}+8\vec{a}-7\vec{b}$
- Compute the norm of the following vectors
 - $\vec{p}=(-10, 19, 32)$
 - $\vec{q}=(24, 50, 17)$
 - $\vec{r}=(4, -16, -27)$

Unit 6: Vector Geometry

The commands for the operations involved in vector geometry are:

Operation	Command
The Dot Product	<code>sum(p.*q)</code>
The Cross Product	<code>cross(p,q)</code>

SciLab Practice

Given $\vec{u} = (1, 5, -2)$, $\vec{v} = (3, -1, 0)$, $\vec{w} = (5, 9, -4)$

- 1) Determine $\vec{u} \cdot \vec{v}$
- 2) Determine $\vec{u} \times \vec{v}$
- 3) Determine whether $\vec{u}$ and $\vec{v}$ are orthogonal.
- 4) Determine whether the three vectors are coplanar

Answers

Chapter	Answers
1.1	1) $x=7, y=2$ 2) $x=-16, y=2$ 3) $x=1, y=1$ 4) $a=\frac{32}{37}, b=\frac{-68}{37}, c=\frac{-7}{37}$ 5) $x=\frac{-3}{7}, y=\frac{2}{7}, z=\frac{8}{7}$ 6) $x=\frac{1}{9}, y=\frac{10}{9}, z=\frac{-7}{3}$

1.2	1) a) $x = \frac{-1}{7}, y = \frac{-3}{7}$ b) $x = \frac{-4}{5}, y = \frac{11}{5}$ c) No solution, inconsistent system d) $x = -21 - 15r, y = -17 - 11r, z = r$ e) No solution, inconsistent system f) $w + y = 0, x - y = 0, z = 0, y = r$ 2) No solution if $b = -2$
2.1	1) $\begin{bmatrix} 2 & 0 & -2 \\ 8 & -18 & 18 \end{bmatrix}$ 2) $\begin{bmatrix} 18 & 25 \\ 3 & -10 \end{bmatrix}$ 3) $\begin{bmatrix} -17 & -6 & 11 \\ 10 & -51 & 15 \end{bmatrix}$ 4) Undefined 5) $\begin{bmatrix} 19 & 25 \\ -1 & 5 \end{bmatrix}$ 6) $\begin{bmatrix} 24 & 20 \\ -6 & 0 \end{bmatrix}$ 7) $\begin{bmatrix} -5 & -1 \\ 0 & -14 \end{bmatrix}$ 8) Undefined 9) $\begin{bmatrix} 1 & 4 \\ 0 & -9 \\ -1 & 9 \end{bmatrix}$ 10) $\begin{bmatrix} 3 & 6 & 3 \\ 0 & 11 & 6 \end{bmatrix}$ 11) $\begin{bmatrix} 10 & -1 \\ 5 & 6 \end{bmatrix}$ 12) 9 13) 24 14) $\begin{bmatrix} 16 & 0 \\ -9 & 25 \end{bmatrix}$

2.2	1) $\begin{bmatrix} -2 & -5 \\ -1 & 4 \end{bmatrix}$ 2) $A(BC) = (AB)C = \begin{bmatrix} 224 & 168 \\ -6 & -12 \end{bmatrix}$ 3) $AB = \begin{bmatrix} 24 & 20 \\ -6 & 0 \end{bmatrix}, BA = \begin{bmatrix} 19 & 25 \\ -1 & 5 \end{bmatrix}$ 4) $\begin{bmatrix} 6 & 5 \\ 0 & 1 \end{bmatrix}$ 5) $\begin{bmatrix} 4 & 0 \\ -1 & 5 \end{bmatrix}$ 6) $\begin{bmatrix} 1 & 2 \\ 10 & 6 \end{bmatrix}$ 7) 14 8) $\begin{bmatrix} 10 & -1 \\ 5 & 6 \end{bmatrix}$ 9) $\begin{bmatrix} 12 & 0 \\ -3 & 15 \end{bmatrix}$ 10) $\begin{bmatrix} 24 & -6 \\ 20 & 0 \end{bmatrix}$
3.1	1) Yes 2) Yes 3) No 4) Yes 5) No 6) a) No b) Yes c) Yes d) No

3.2	1) $\begin{bmatrix} \frac{1}{3} & 0 \\ \frac{1}{12} & \frac{-1}{4} \end{bmatrix}$ 2) $\begin{bmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{-3}{5} & \frac{4}{5} \end{bmatrix}$ 3) No inverse 4) $A^{-1} = \begin{bmatrix} 0 & -1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, A^{-2} = \begin{bmatrix} \frac{-1}{2} & \frac{-1}{2} \\ \frac{1}{4} & \frac{-1}{4} \end{bmatrix}$ 5) $\begin{bmatrix} \frac{1}{11} & \frac{1}{11} \\ \frac{3}{11} & \frac{-8}{11} \end{bmatrix}$ 6) $\begin{bmatrix} -8 & 1 \\ 9 & -1 \end{bmatrix}$ 7) $\begin{bmatrix} 1 & 9 \\ 1 & 8 \end{bmatrix}$ 8) No inverse 9) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ will have no inverse because $ad - bc = 0$ and there is no matrix multiplied by $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ will result in a product of I
3.3	1) $\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{10} & \frac{-1}{10} & \frac{1}{5} \\ \frac{-7}{2} & \frac{1}{2} & 0 \end{bmatrix}$

	$2) \begin{bmatrix} \frac{1}{10} & \frac{2}{5} & \frac{-1}{10} \\ \frac{-1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{-9}{10} & \frac{7}{5} & \frac{-1}{10} \end{bmatrix}$ 3) No inverse $4) \begin{bmatrix} \frac{-3}{2} & \frac{-3}{2} & \frac{11}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{-3}{2} \\ \frac{1}{2} & \frac{-1}{2} & \frac{-1}{2} \end{bmatrix}$ 5) No inverse $6) \begin{bmatrix} \frac{1}{2} & 0 & \frac{-1}{2} \\ \frac{-5}{4} & \frac{1}{2} & \frac{5}{4} \\ \frac{-3}{4} & \frac{1}{2} & \frac{-1}{4} \end{bmatrix}$ $7) \begin{bmatrix} -2 & 3 & -1 \\ \frac{4}{5} & \frac{-4}{5} & \frac{1}{5} \\ \frac{9}{5} & \frac{-14}{5} & \frac{6}{5} \end{bmatrix}$
3.4	1) $x=5, y=-1$ 2) $x=5, y=9$ 3) $x=2, y=3$ 4) $x=\frac{13}{5}-\frac{6}{5}t, y=t$ 5) Inverse does not exist 6) Inverse does not exist 7) $x=\frac{-20}{17}, y=\frac{12}{17}, z=\frac{59}{17}$ 8) $x=127, y=-95, z=-38$

4.1	1) a) B, C, D, E, F, G, H, J b) B, C, D, H, J c) C, H, J d) C, E, F, G, H, J e) C, F, G, H, J f) C, E 2) $\begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & -\frac{1}{6} \end{bmatrix}$ 3) $\begin{bmatrix} 144 & 0 & 0 \\ 0 & 1296 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 4) $\begin{bmatrix} \frac{1}{8} & 0 & 0 \\ -\frac{7}{8} & 1 & 0 \\ -\frac{9}{14} & \frac{5}{7} & \frac{1}{7} \end{bmatrix}$ 5) $\begin{bmatrix} 14 & 1 & 5 \\ 1 & 10 & 4 \\ 5 & 4 & -6 \end{bmatrix}, \text{ also symmetric}$ 6) $\begin{bmatrix} 8 & 0 & 0 \\ 4 & 9 & 0 \\ 5 & 2 & -1 \end{bmatrix}, \text{ lower triangle matrix}$ 7) a) True b) False c) False d) False
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4.2	1) a) -36 b) -1 c) 0 d) -9 2) a) $\begin{bmatrix} 1 & -5 \\ 0 & 6 \end{bmatrix}$ b) $\begin{bmatrix} 1 & -2 \\ 8 & -1 \end{bmatrix}$ c) $\begin{bmatrix} -11 & 5 & 1 \\ -1 & 8 & -15 \\ 5 & -40 & -8 \end{bmatrix}$ d) $\begin{bmatrix} 15 & -6 & -51 \\ 36 & -42 & -12 \\ -99 & 12 & 33 \end{bmatrix}$ e) $\begin{bmatrix} -25 & -23 & 33 \\ -35 & -2 & 16 \\ 47 & 7 & -56 \end{bmatrix}$
4.3	1) a) 2 b) 48 2) a) -3 b) -3 c) -3 d) -3 3) a) -26 b) 161 c) 94
4.4	1) $x=5, y=-1$ 2) $x=5, y=9$ 3) $x=2, y=3$ 4) No solution 5) $x=\frac{-20}{17}, y=\frac{12}{17}, z=\frac{59}{17}$ 6) $x=127, y=-95, z=-38$

5.1	1) $\overrightarrow{MN} = (9, 0, -18), \overrightarrow{NM} = (-9, 0, 18)$ 2) $\vec{u} = 3\vec{v}$, parallel 3) $\vec{u} = \frac{6}{5}\vec{w}$, parallel 4) $\vec{u} = -\vec{w}$, yes 5) $\vec{u} \neq -\vec{w}$, no 6) $(4, -9, -8)$ 7) a) $(3, 9, 38)$ b) $(-64, 15, 72)$ c) $(-2, -32, -16)$ d) $(24, 7, -12)$ e) $(-119, 55, 140)$
5.2	1) a) $\left(\frac{6}{\sqrt{125}}, \frac{5}{\sqrt{125}}, \frac{8}{\sqrt{125}}\right)$ b) $\left(\frac{3}{\sqrt{155}}, 0, -\frac{11}{\sqrt{155}}, \frac{5}{\sqrt{155}}\right)$ c) $(0, 1, 0, 0, 0, 0)$ d) $\left(\frac{-3}{\sqrt{314}}, \frac{-17}{\sqrt{314}}, \frac{-4}{\sqrt{314}}\right)$ 2) Yes 3) a) $5\hat{i} - 7\hat{j} - 9\hat{k}$ b) $1(1, 0, 0, 0, 0) + 2(0, 0, 1, 0, 0) + 5(0, 0, 0, 1, 0) - 12(0, 0, 0, 0, 1)$ c) $-4(1, 0, 0, 0) + (0, 1, 0, 0) + 9(0, 0, 1, 0) - 3(0, 0, 0, 1)$ 4) a) $\sqrt{128} \hat{i} 11.31$ b) $\sqrt{46} \hat{i} 6.78$ c) $\sqrt{12} \hat{i} 3.46$

6.1	1) a) 9 b) -28 c) $\dot{i} 5.85$ 2) a) Yes b) Yes c) No 3) $\cos 0^\circ = 1$, so $\vec{u} \cdot \vec{v} = \vec{u} \vec{v}$ 4) $y = 5$ 5) 64° 6) 513 7) $\vec{p} \cdot \vec{q} = \vec{q} \cdot \vec{p} = 9$
6.2	1) a) Yes b) No c) No 2) Orthogonal, but not orthonormal 3) $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)$ 4) $x = \frac{-5}{2}$ 5) a) $\left(\frac{6}{7}, -\frac{3}{7}, \frac{9}{7}\right)$ d) $\left(\frac{22}{9}, \frac{11}{18}, \frac{11}{18}\right)$ e) $\left(\frac{270}{203}, -\frac{855}{203}, -\frac{135}{203}\right)$
6.3	1) a) $(100, -20, 12)$ b) $(50, 12, 21)$ 2) a) $(-111, 49, -16)$ b) $(2, 14, -97)$ 3) a) $\sqrt{1301} \dot{i} 36.07$ b) $\sqrt{3659} \dot{i} 60.49$

	4) $\sqrt{5100} \approx 71.41$ 5) 77.26° 6) a) $(7, 26, 24) = -(-7, -26, -24)$ b) $(0, 0, 0)$
6.4	1) $\vec{u} \cdot (\vec{v} \times \vec{w}) = -76$ 2) a) Yes b) No c) No 3) $\vec{d} \cdot (\vec{c} \times \vec{f}) = -4 + 10y - 60 = 0$ $y = 6.4$ 4) $\vec{a} \cdot (\vec{b} \times \vec{c}) = 3x - 60 - 6 = 0$ $x = 22$ 5) $x = \frac{13}{3}$ 6) a) -214 b) $(332, -15, -6)$